

Introduction to jet quenching phenomenology in the quark-gluon plasma

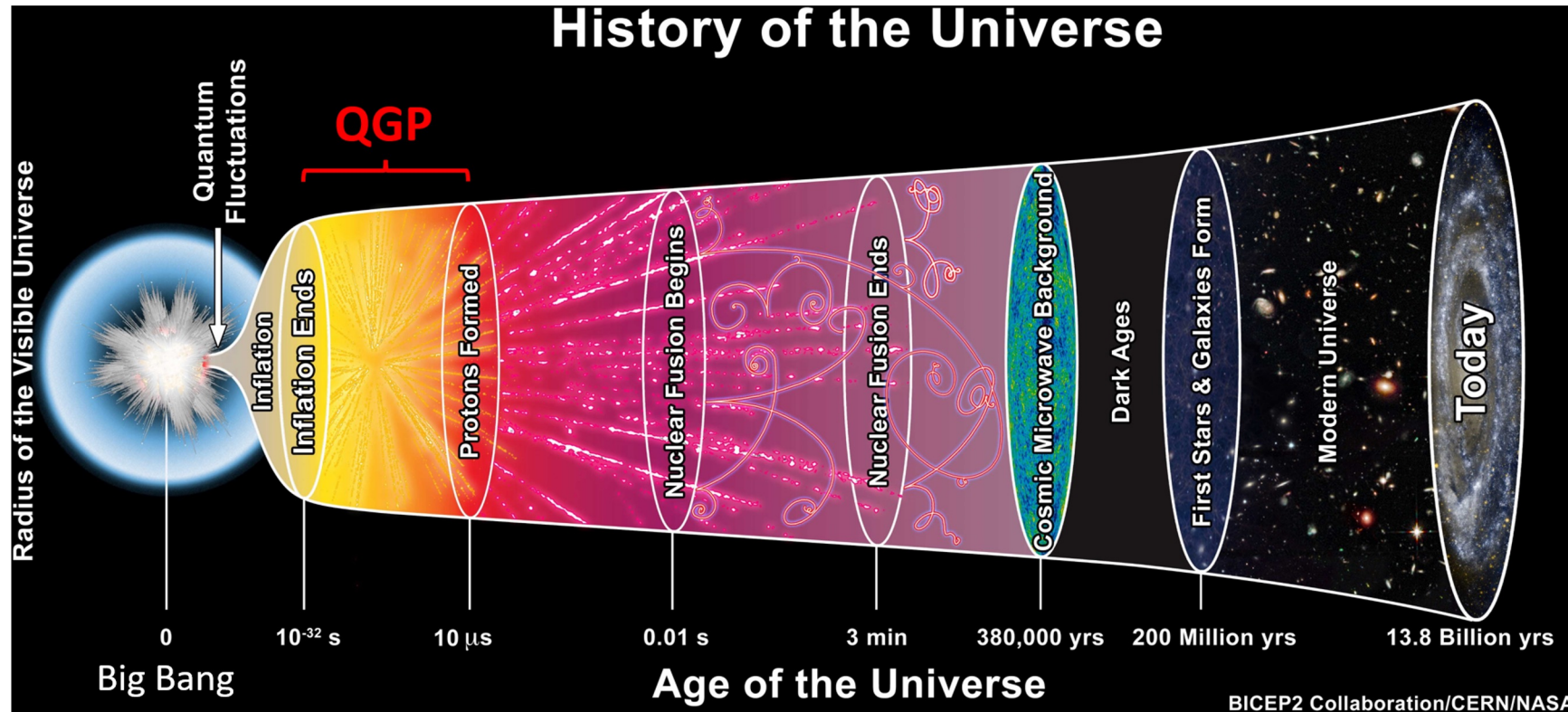
Coleridge Faraday

University of Cape Town, South Africa

19th August 2026, CTMP seminar 1

Based on **CF**, W. A. Horowitz, B. Schenke [arXiv:2608.xxxxx](#),
B. Bert, **CF**, and W. A. Horowitz, [arXiv:2603.21861](#),
CF, B. Bert, J. Brand, W. Vogelsang, and W. A. Horowitz, [arXiv:2512.17832](#)
and **CF** and W. A. Horowitz, [JHEP 11, 019 \(2025\)](#)

The Quark-Gluon Plasma

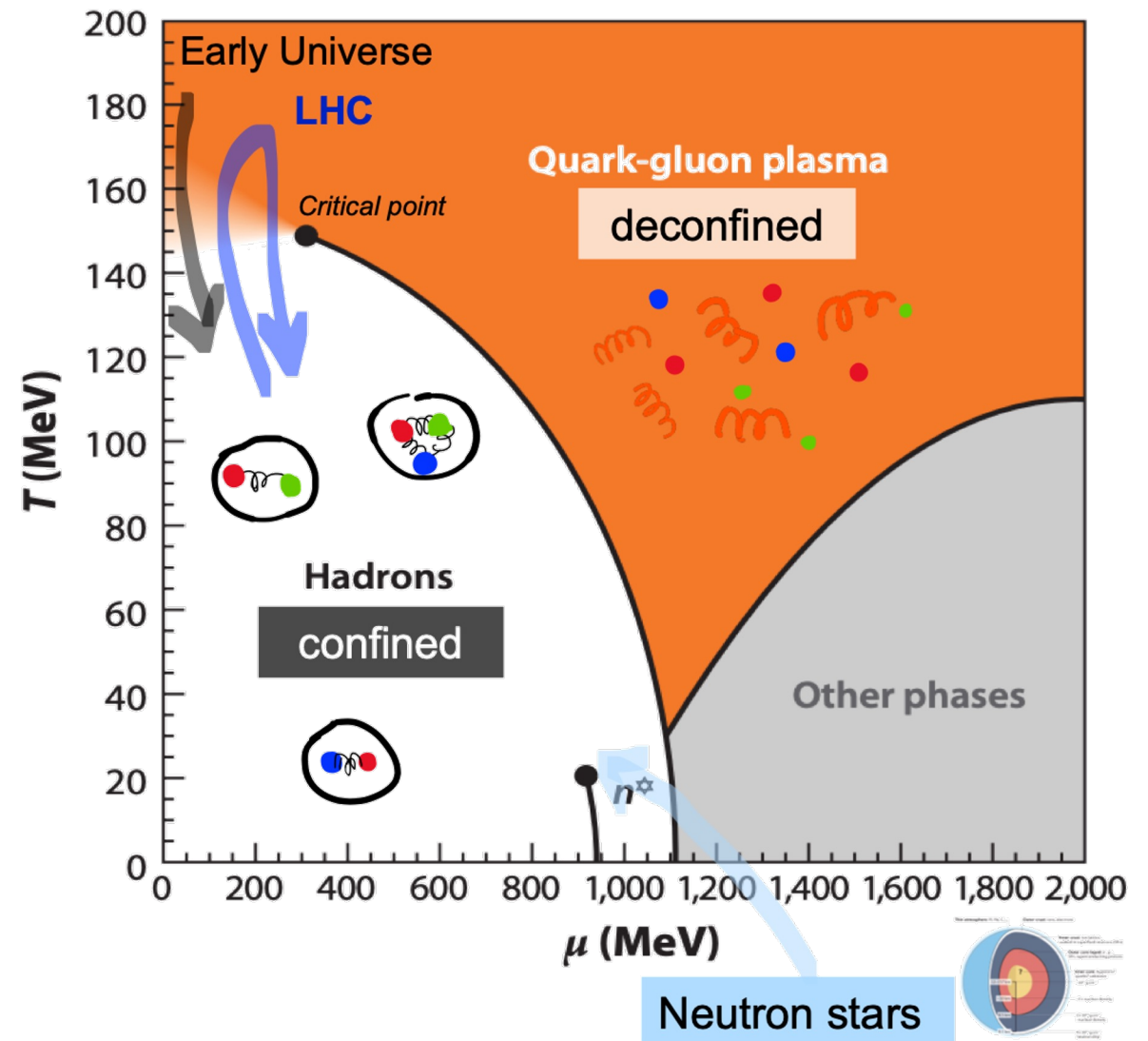


The early universe was made up of a dense, hot soup of quarks and gluons: the **quark-gluon plasma**

QCD Matter at the Extreme

- High energy collisions of heavy-ions probe the *high temperature* regime of the QCD phase diagram
- QGP formation established in collision of large nuclei

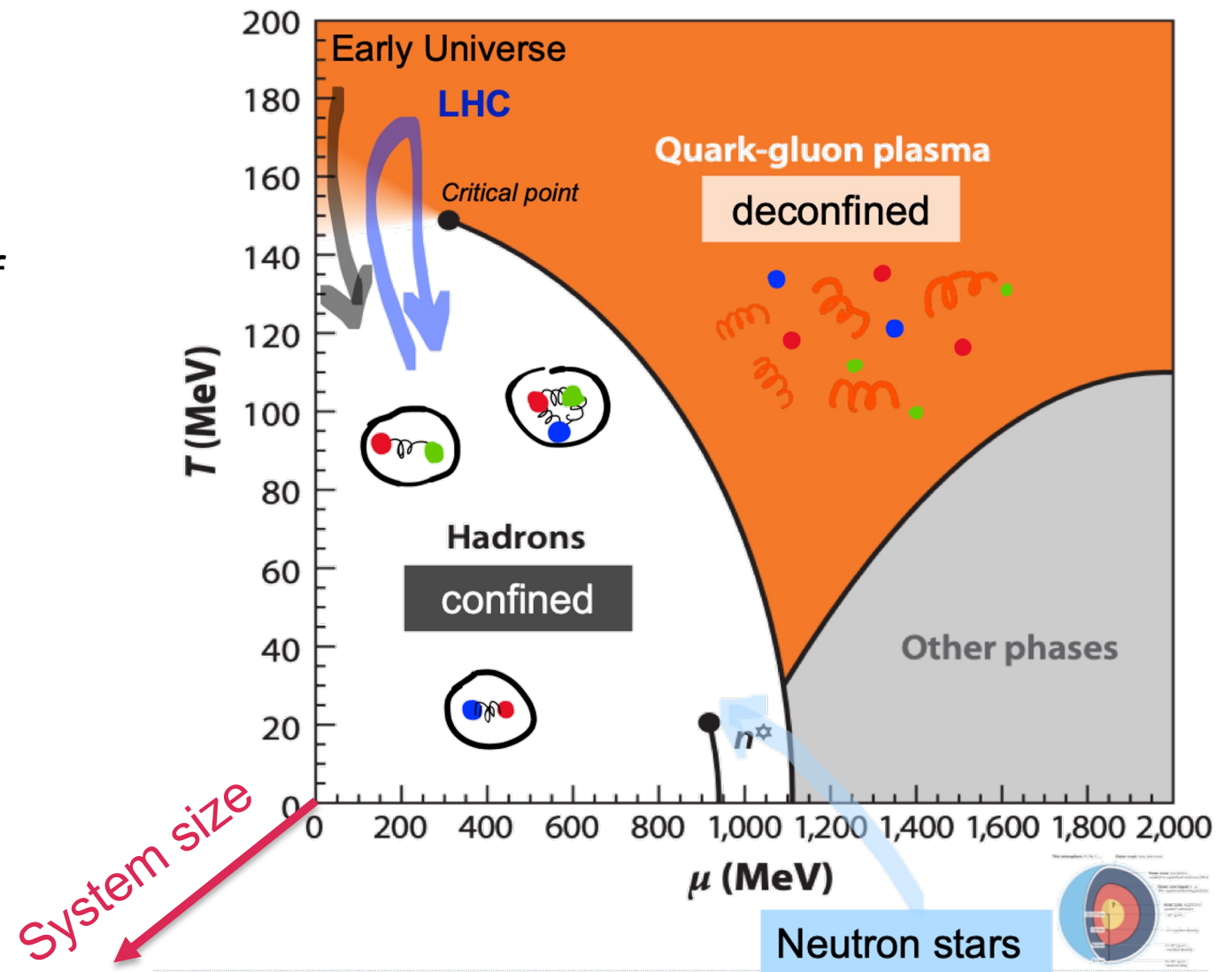
Image:
https://opendata.atlas.cern/docs/documentation/introduction/heavy_ions



QCD Matter at the Extreme

- High energy collisions of heavy-ions probe the *high temperature* regime of the QCD phase diagram
- QGP formation established in collision of large nuclei
- New frontier: **how small can a collision system be and still form a collective, nearly-perfect fluid?**

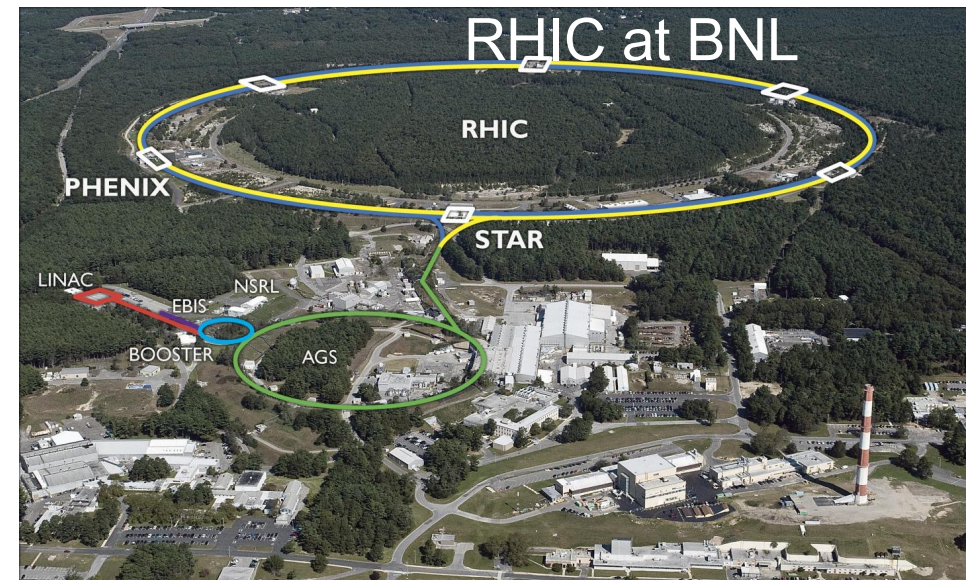
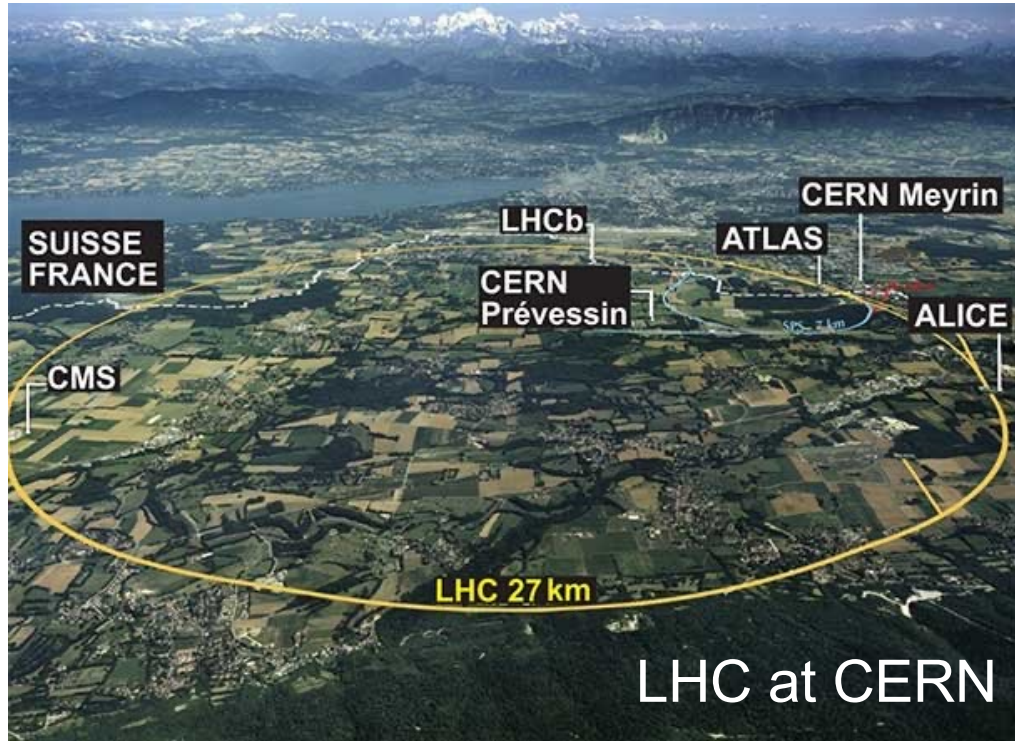
Image:
https://opendata.atlas.cern/docs/documentation/introduction/heavy_ions



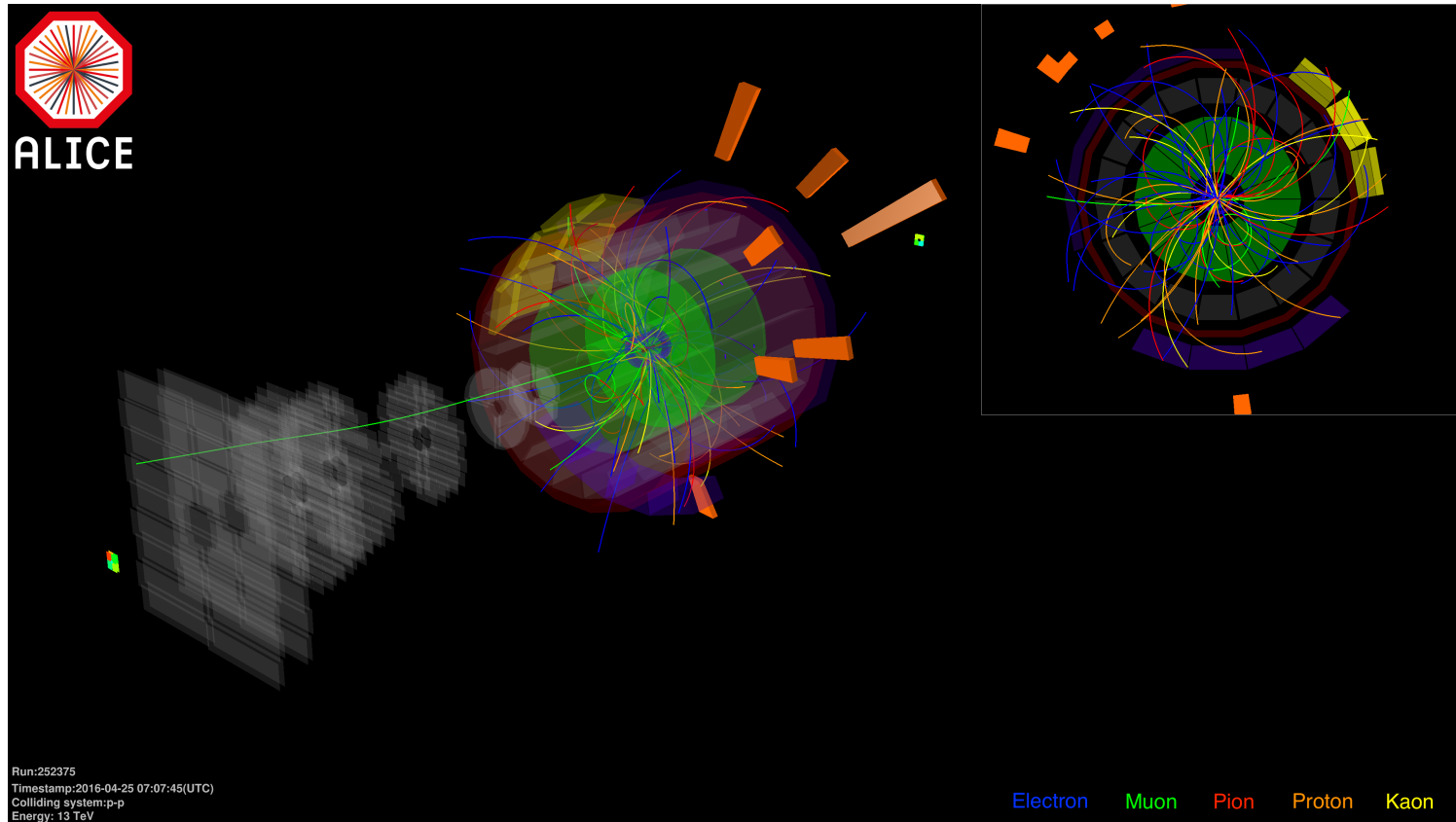
> Large-hadron colliders



>99.9999% of the speed of light



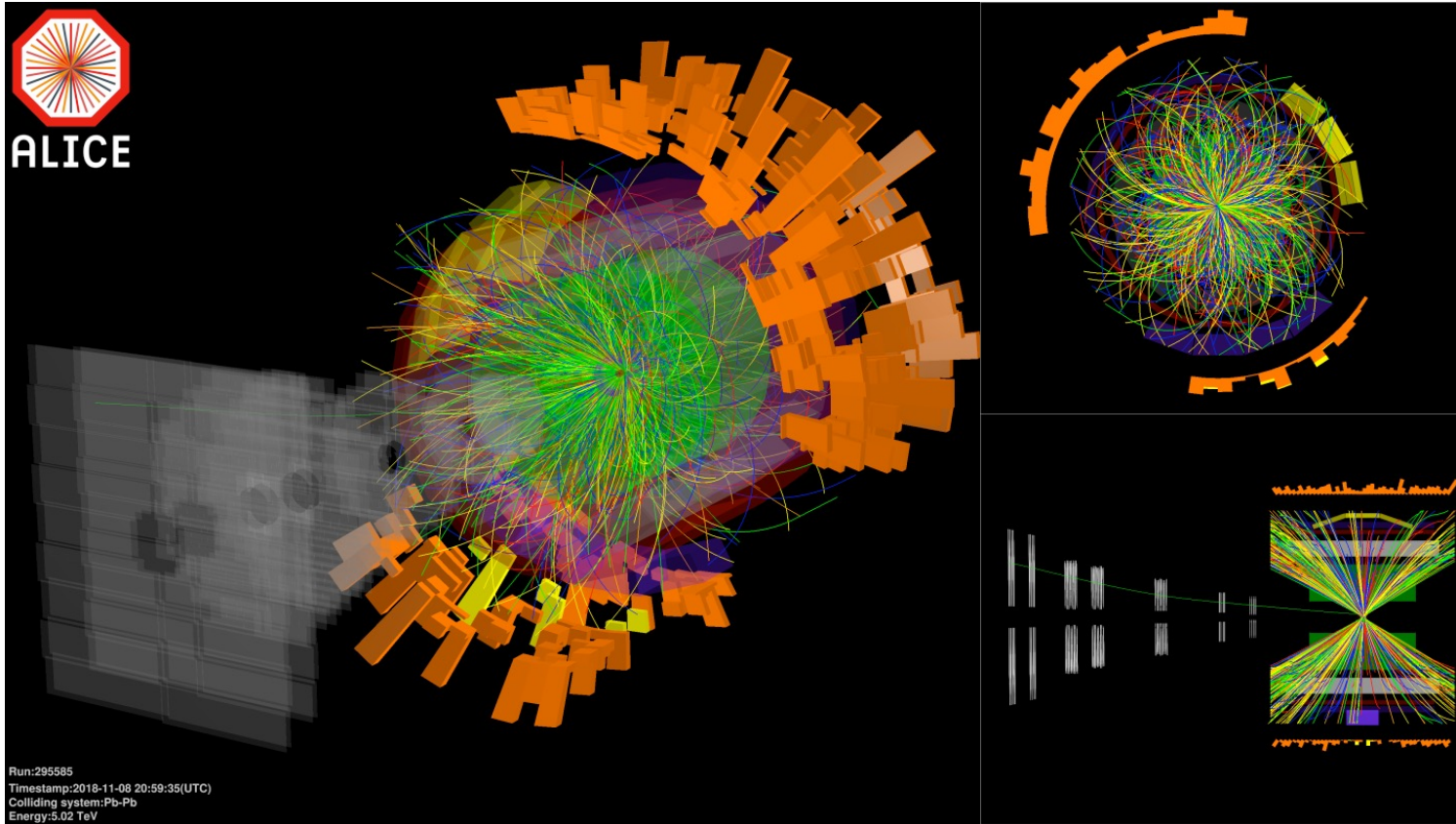
> p + p collisions



- Teaches us about the standard model
- Potential way to find beyond the standard model (BSM) physics
- Used as a baseline for understanding heavy-ion collisions

Source: CERN, <https://cds.cern.ch/record/2149032>

Heavy-ion collisions: what we see



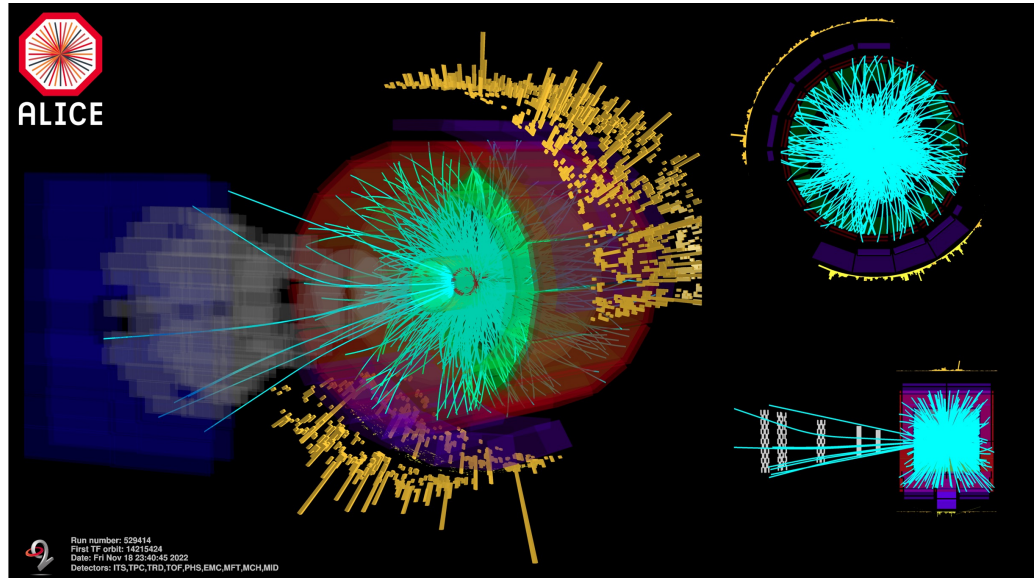
- **Many, many more particles**
- Information from the collision is encoded in the properties of the final state particles
- Observables must be constructed which are sensitive to various properties of the state of matter formed in the collision

Source: CERN, <https://home.cern/news/news/accelerators/time-lead-collisions-lhc>

Measuring the quark-gluon plasma

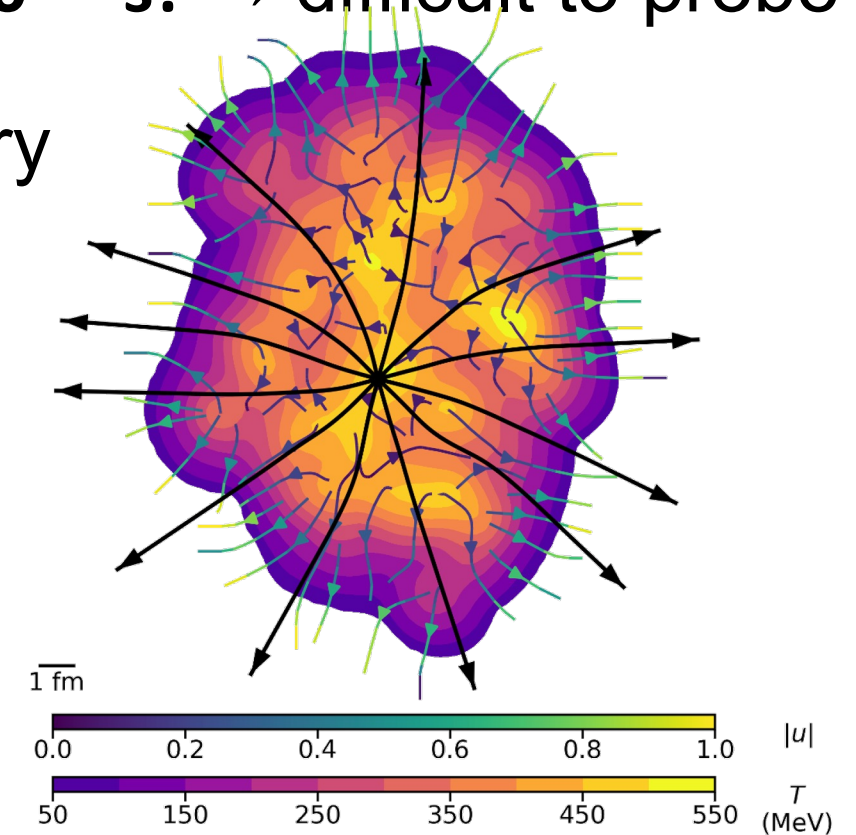
QGP in heavy-ion collisions lasts for only $\simeq 10^{-23}$ s! $\rightarrow$ difficult to probe

Experiment



Source: CERN

Theory

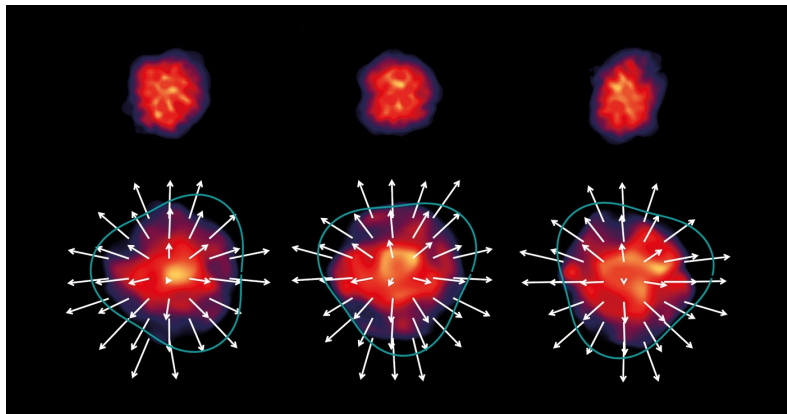


J. Bahder, H. Rahman, M. D. Sievert, and I. Vitev, arxiv: 2412.05474

Signatures of the QGP

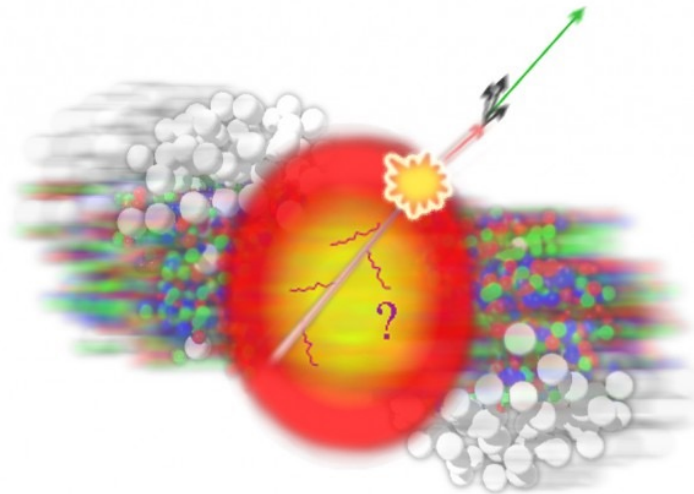
Collective Flow

Pressure gradients change the distribution of particles



Jet Quenching

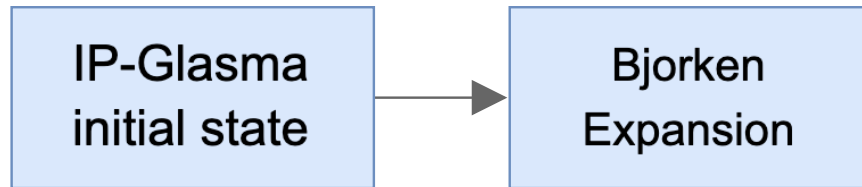
Particles lose energy through interactions in the QGP



And many, many more; but not enough time to discuss today!

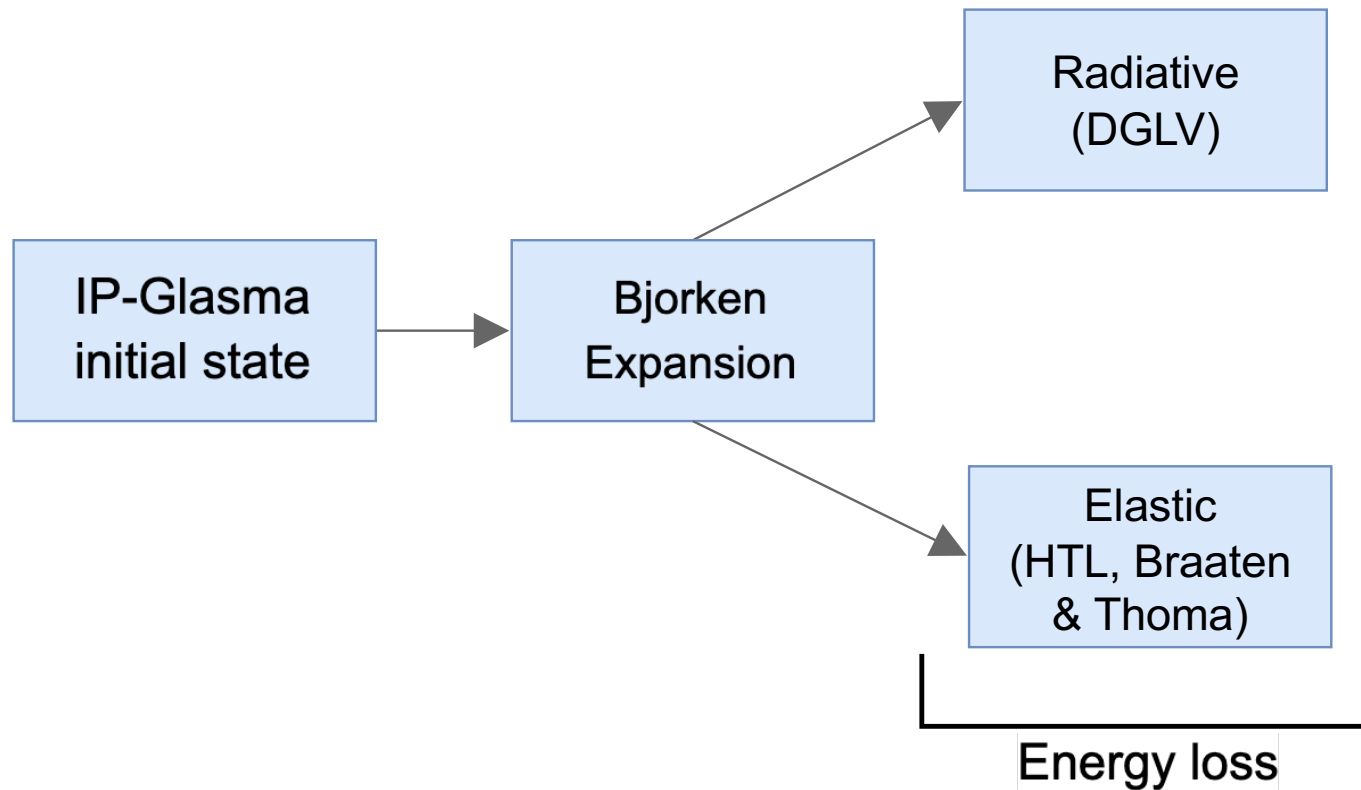
> The “Cape Town” model

And more generally, the typical structure
of jet quenching models



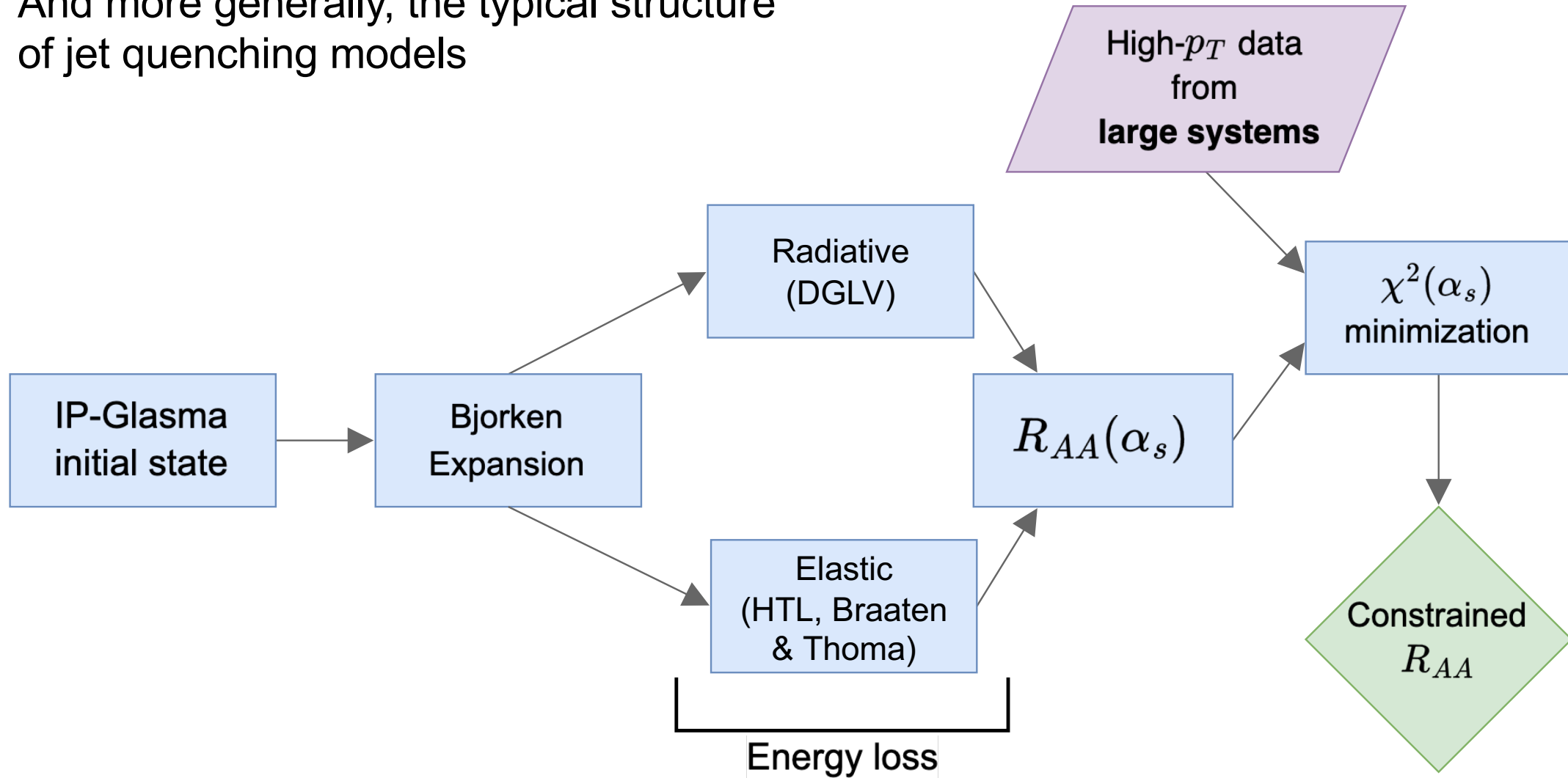
The “Cape Town” model

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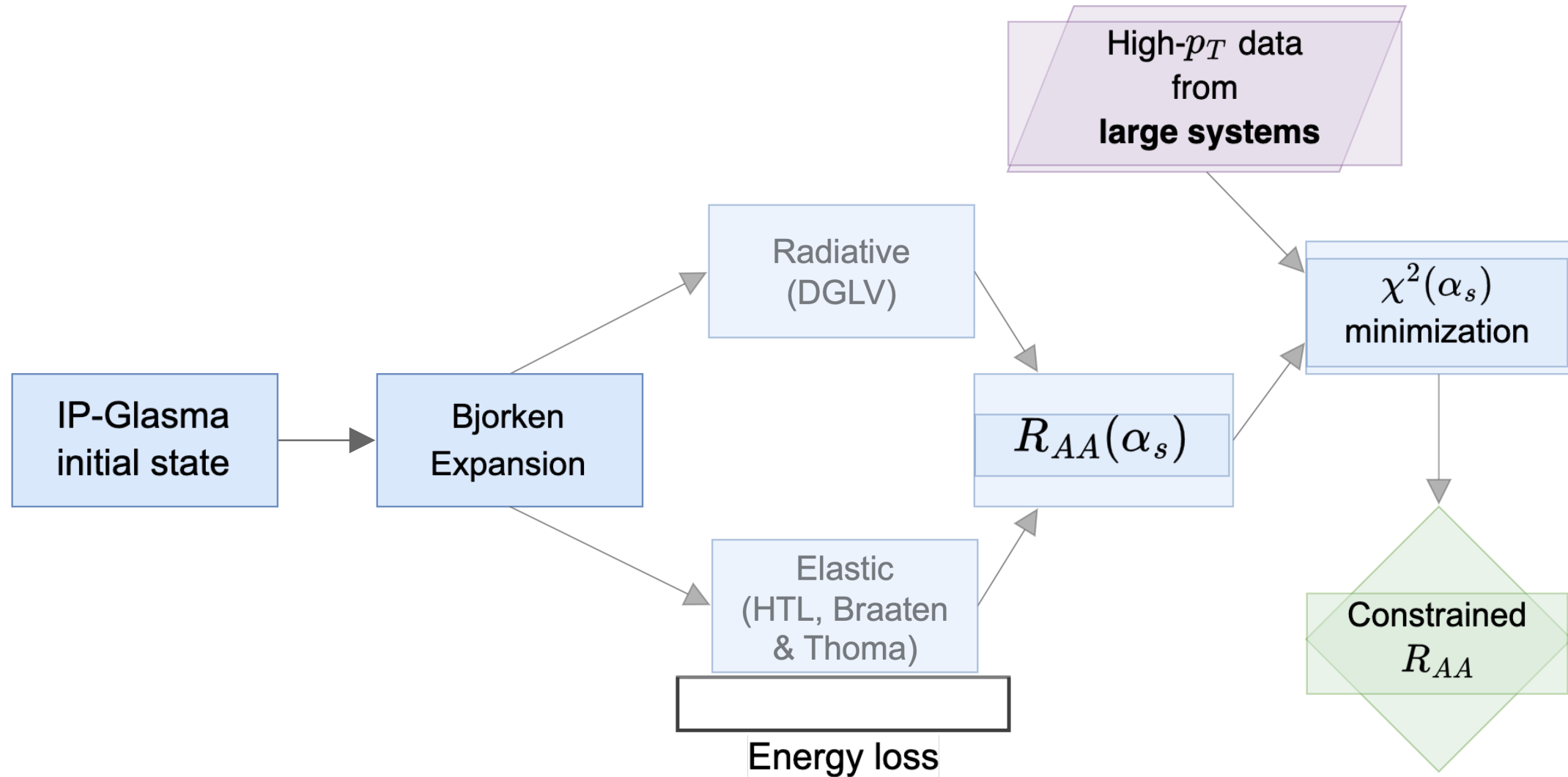


The “Cape Town” model

And more generally, the typical structure of jet quenching models



Hydrodynamics and geometry



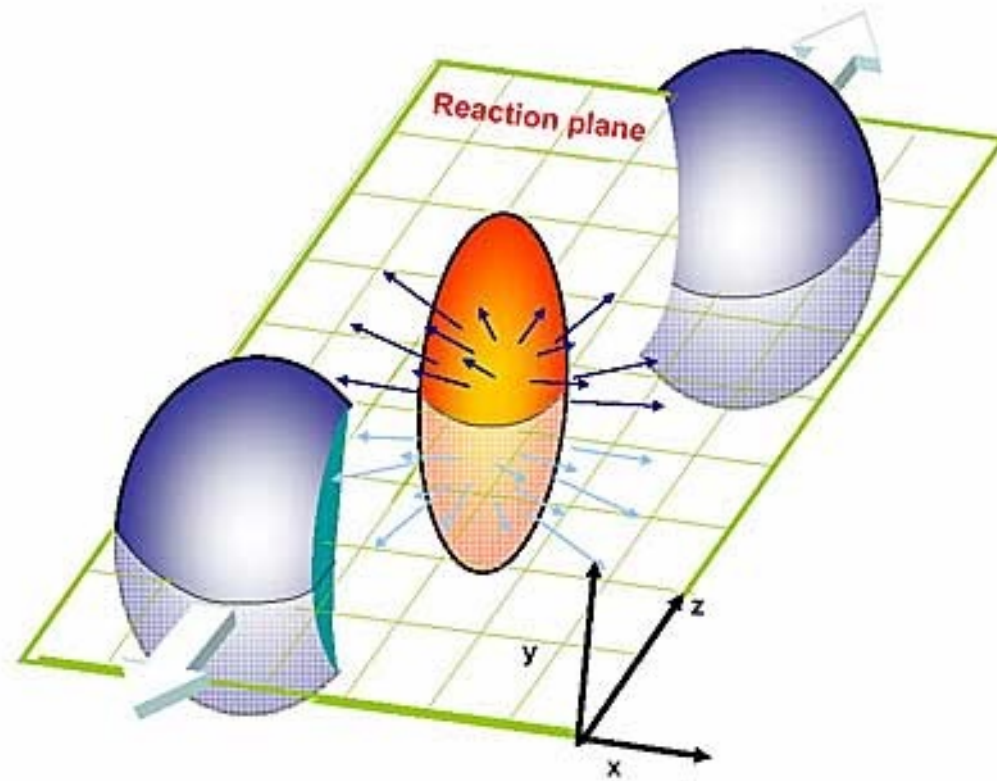


The Glauber Model

> The Glauber model

The *Glauber Model* is used to understand **geometrical aspects** of a heavy-ion collision

- Assumes
 - Nucleons are **distributed independently** in the nucleus
 - Nucleons interact with other nucleons, but are **undeflected**



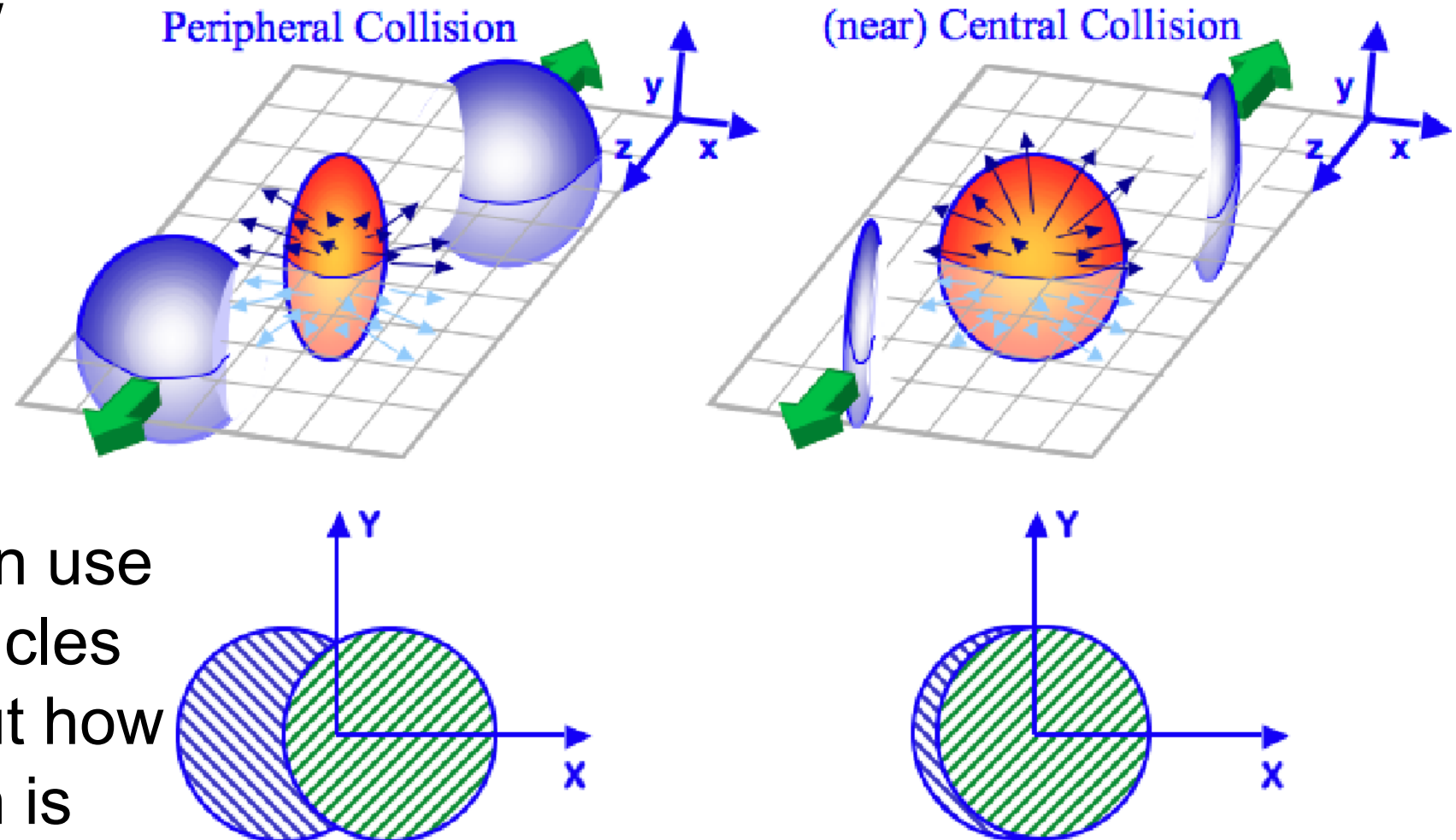
M. L. Miller, K. Reygers, S. J. Sanders, and P. Steinberg, Ann. Rev. Nucl. Part. Sci. 57, 205 (2007).

Centrality

One talks about how *central* a collision is

The more the two nuclei overlap, the more central

Experimentalists can use the number of particles produced to work out how central a collision is

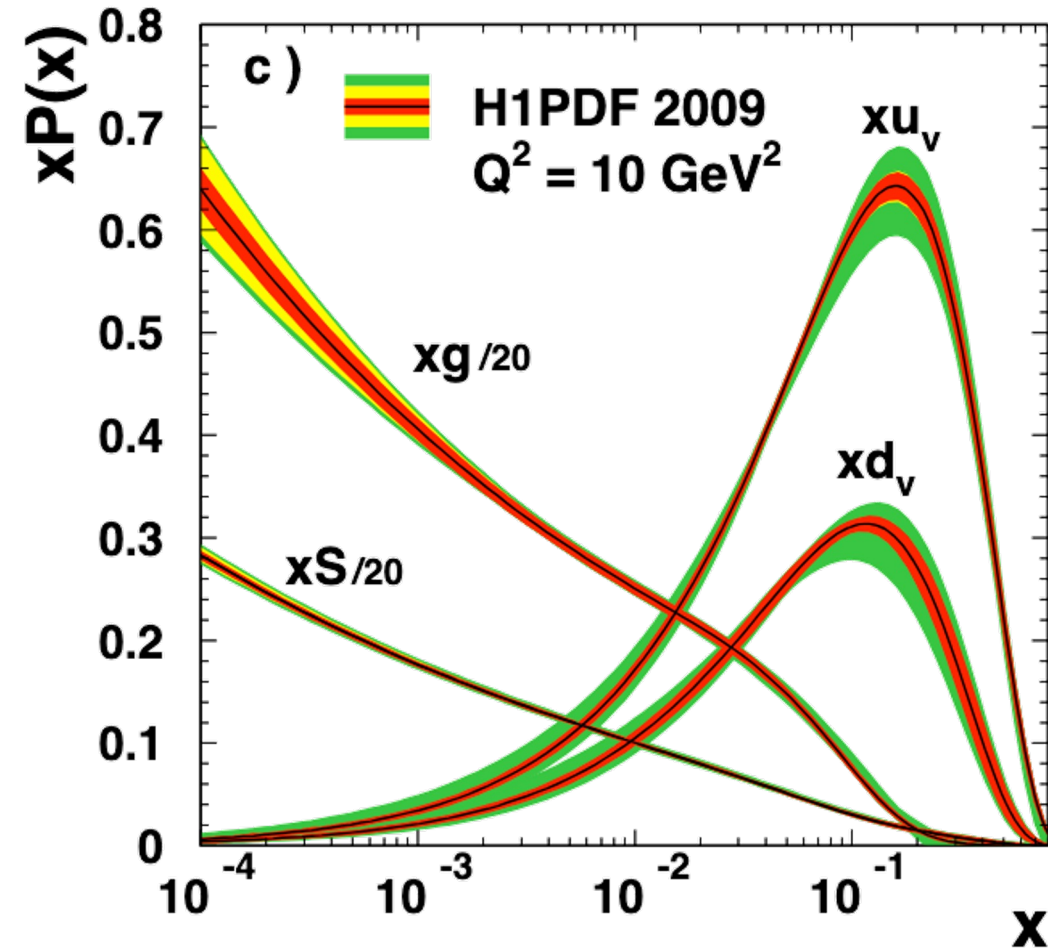




Saturation and the initial state

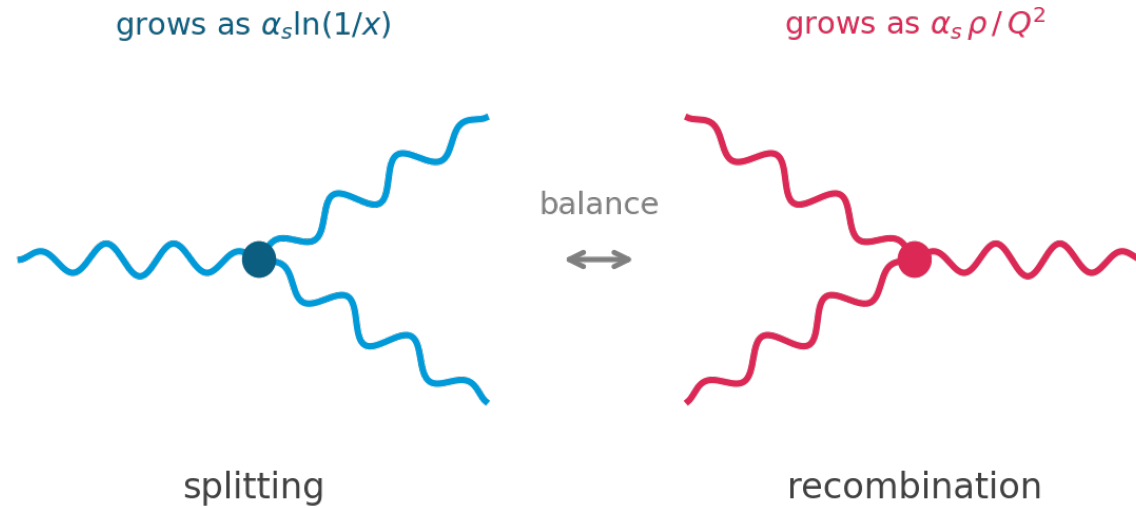
> What are we actually colliding?

- A **10 GeV** probe at mid-rapidity samples the proton at $x \sim 10^{-3}$, deep in the small- x region
- The gluon density rises steeply, roughly as a power of $1/x$
 - Hard production at the LHC is overwhelmingly gluonic
- Can this growth continue indefinitely?

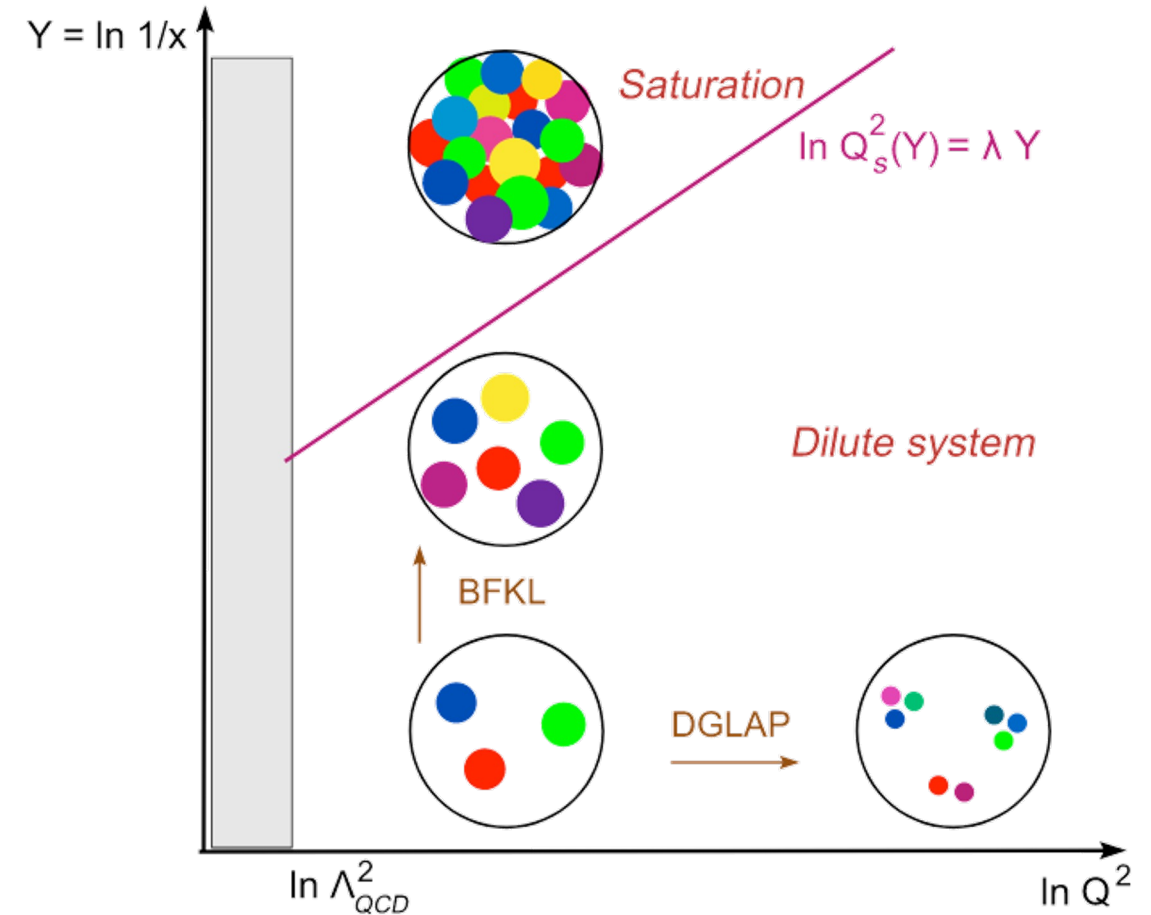


F. Gelis, E. Iancu, J. Jalilian-Marian, and R. Venugopalan, *Ann. Rev. Nucl. Part. Sci.* **60**, 463 (2010).

> Gluon saturation



- Recombination balances splitting: occupation number saturates at $\sim 1/\alpha_s$
- An emergent **saturation scale** $Q_s(x, b, A)$, growing with nuclear size and with $1/x$
- For Pb at the LHC, $Q_s \sim 2$ GeV

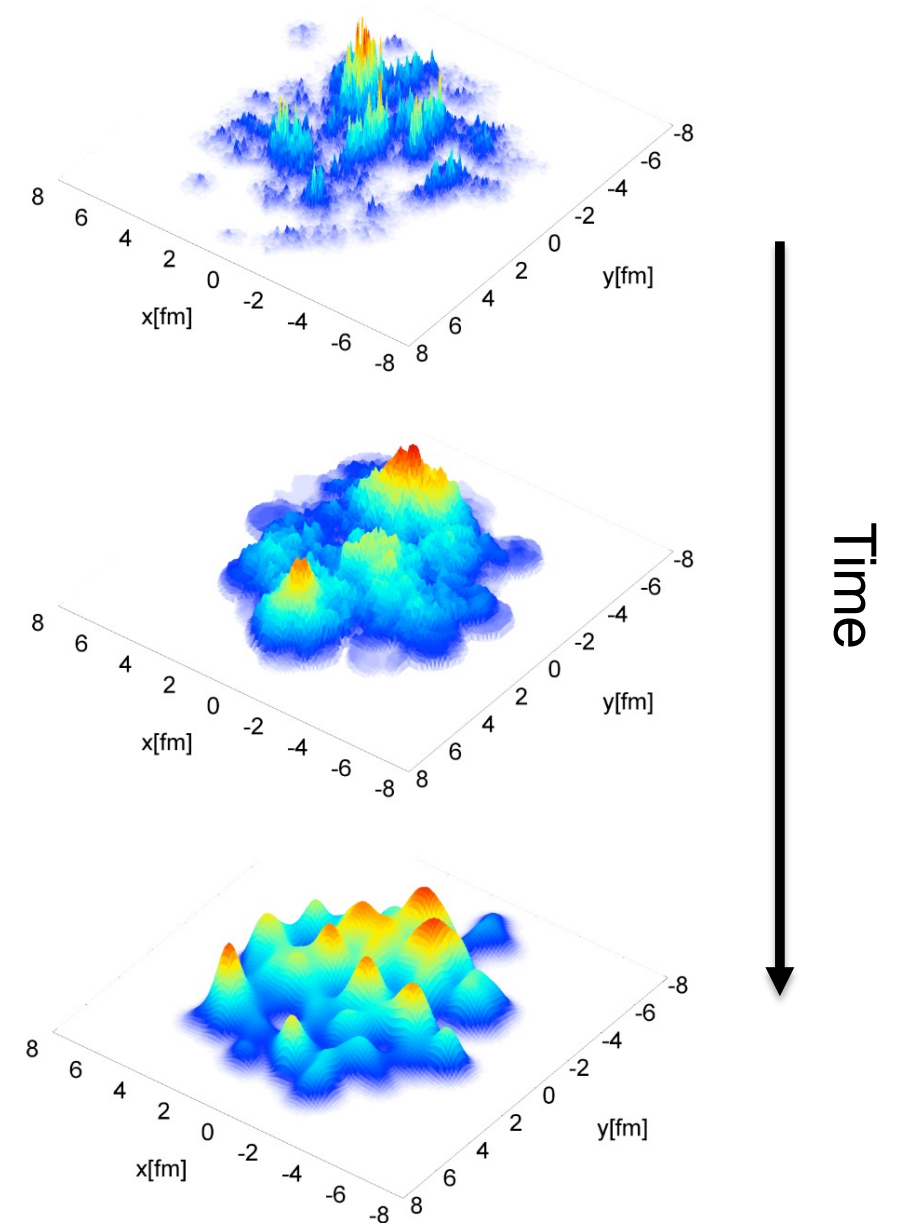


F. Gelis, E. Iancu, J. Jalilian-Marian, and R. Venugopalan, *Ann. Rev. Nucl. Part. Sci.* **60**, 463 (2010).

IP-Glasma

- Impact-parameter dependent saturation, fitted to HERA DIS data
- Event by event: sample nucleon positions, sample colour charges, evolve the classical fields
- Gives the lumpy initial energy density that seeds the hydrodynamics
 - The same framework behind the flow comparison coming up
- Evolved according to classical Yang Mills
- One framework produces **both** the soft bulk and the semi-hard particles
 - An alternative to the simpler Glauber-model-based ideas

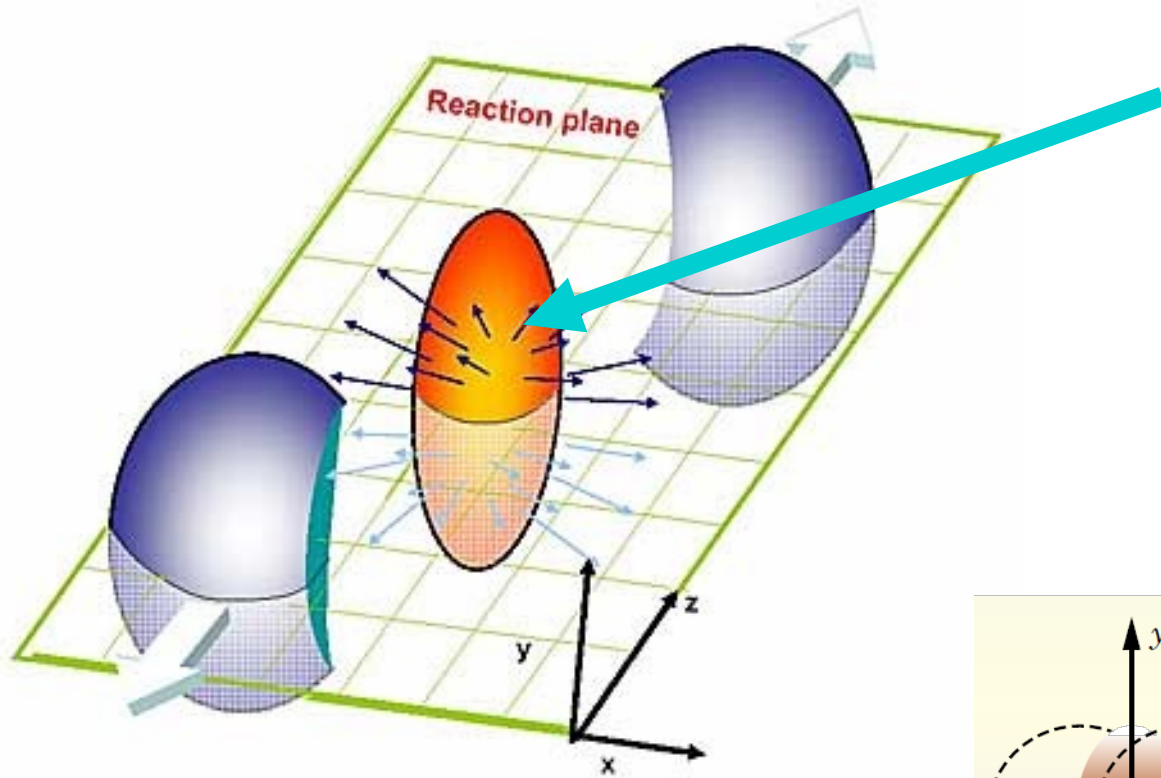
Mäntysaari and Schenke, *Phys. Rev. D* 94, 034042 (2016);
Schenke, Tribedy and Venugopalan, *PRL* 108, 252301 (2012)



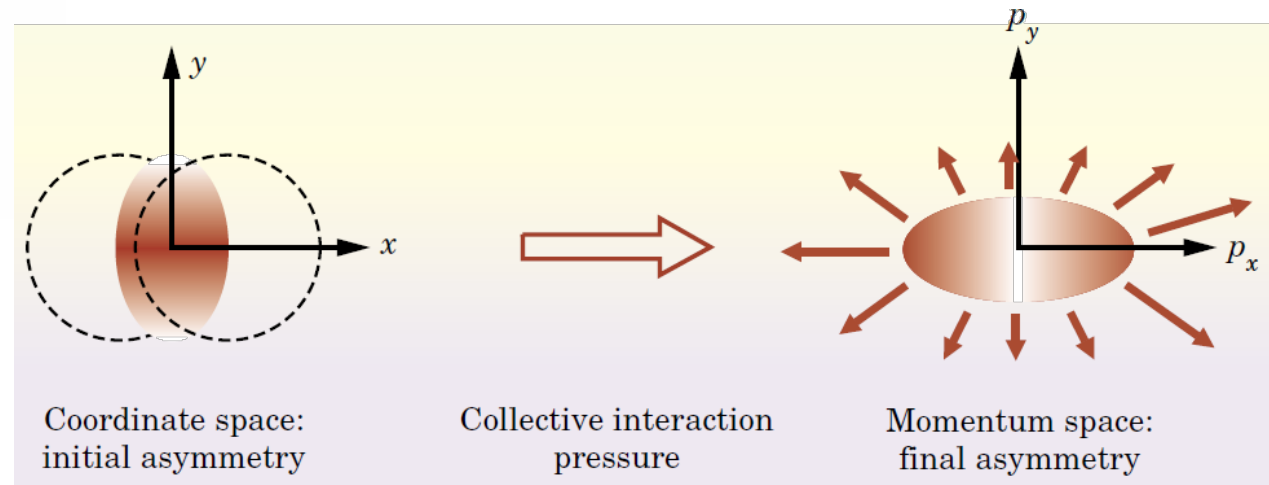


Collective flow

Collective flow



Non-symmetrical shape initially
 $\Rightarrow$
Non-symmetric final state in
momentum space, due to
pressure gradients

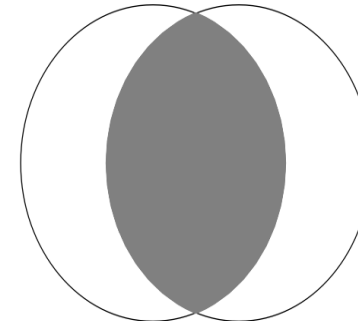
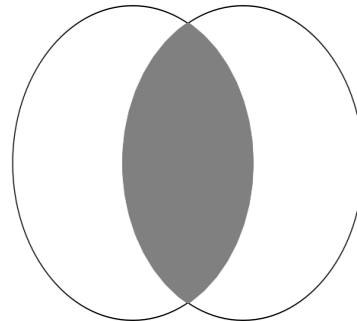
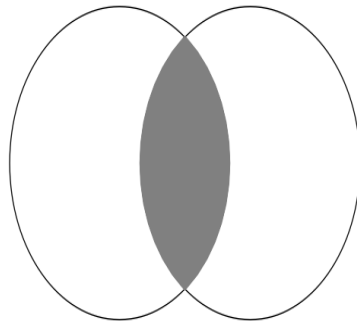
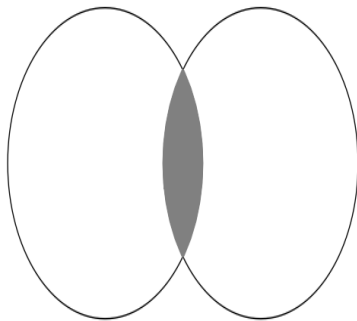


> Elliptic flow and v_2

- The v_2 captures the *ellipticity* of the collision

- v_2 is the Fourier coefficient of $\cos(2 \varphi)$

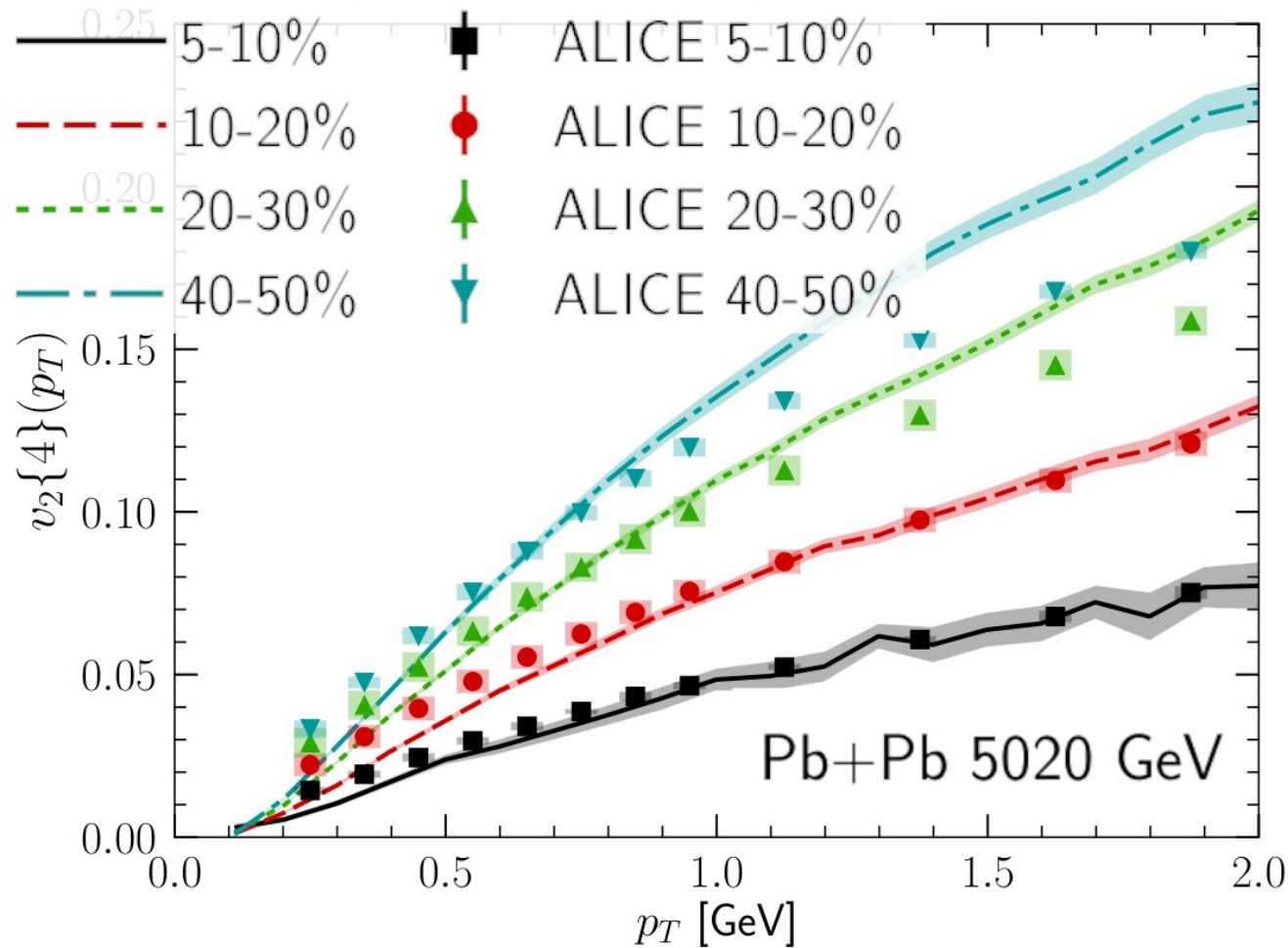
Very elliptic



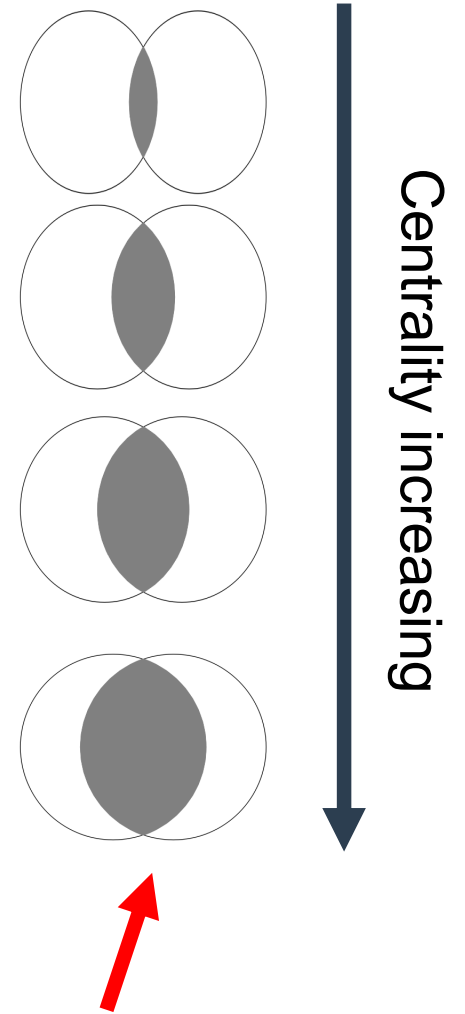
Less elliptic

Quantitative comparison to data

Hydrodynamics
models describe
data



Schenke et al., Phys. Rev. C 102 (2020) 044905



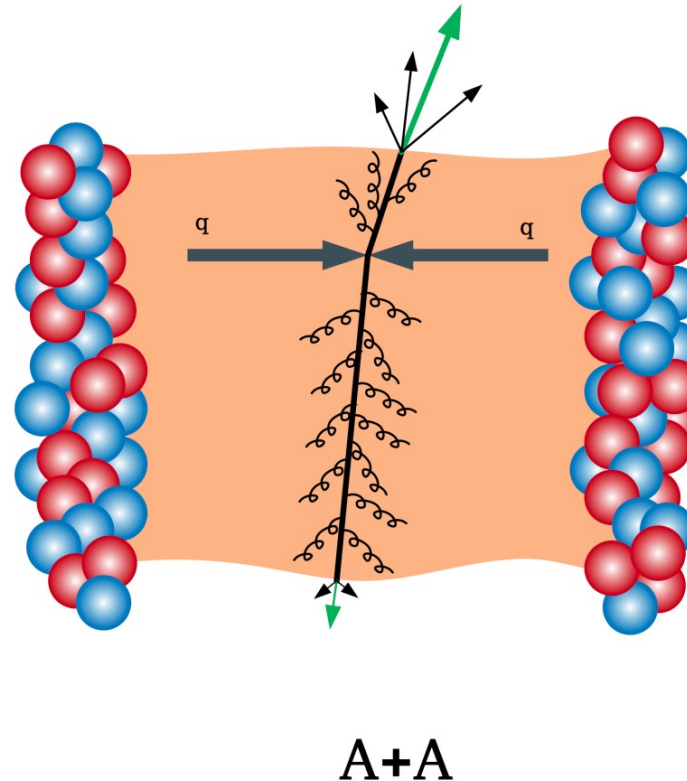
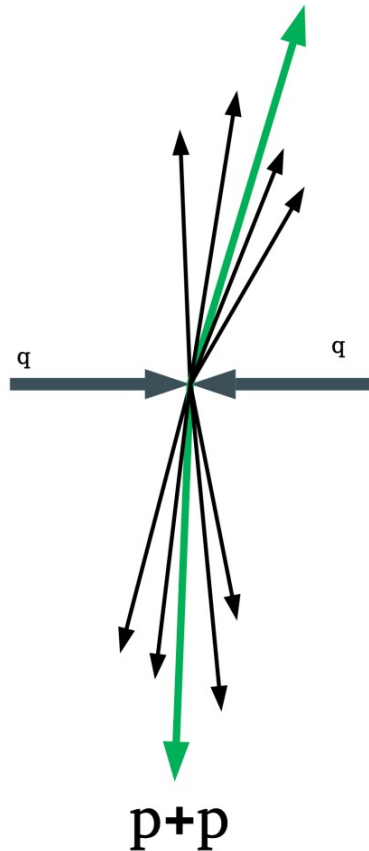
Less elliptic, less flow



Jet quenching

Energy loss

- As quarks and gluons move through the plasma, they lose energy

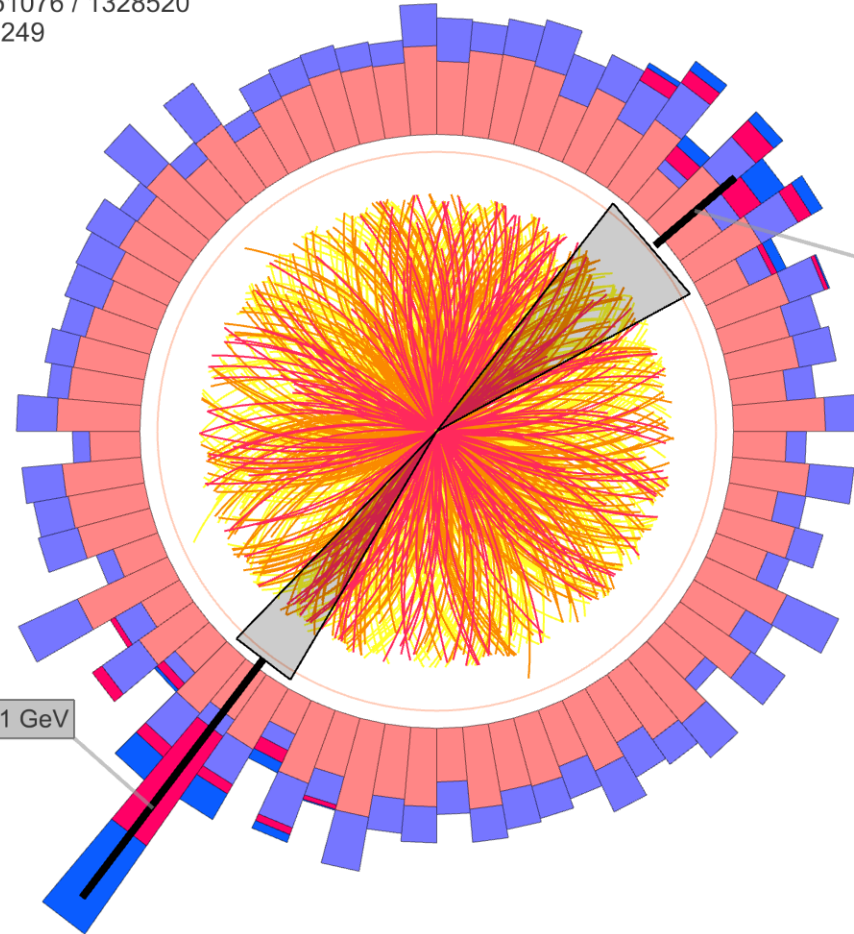


<https://atlas.cern/Updates/Feature/Heavy-Ion-Physics>

Seeing energy loss



CMS Experiment at LHC, CERN
Data recorded: Sun Nov 14 19:31:39 2010 CEST
Run/Event: 151076 / 1328520
Lumi section: 249



Energy = 70GeV

Jet 1, pt: 70.0 GeV

Energy = 200GeV

Jet 2, pt: 205.1 GeV

Where did the
energy go?
A: Dissipated
into the QGP!

<https://cms.cern/news/jet-quenching-observed-cms-heavy-ion-collisions>

> How many particles are produced?

Baseline (pp collisions):

$$\frac{d\sigma^{pp}}{dp_T} = f_p \otimes f_p \otimes \frac{d\hat{\sigma}}{d\hat{t}} \otimes D_{h/c}$$

Measured cross section

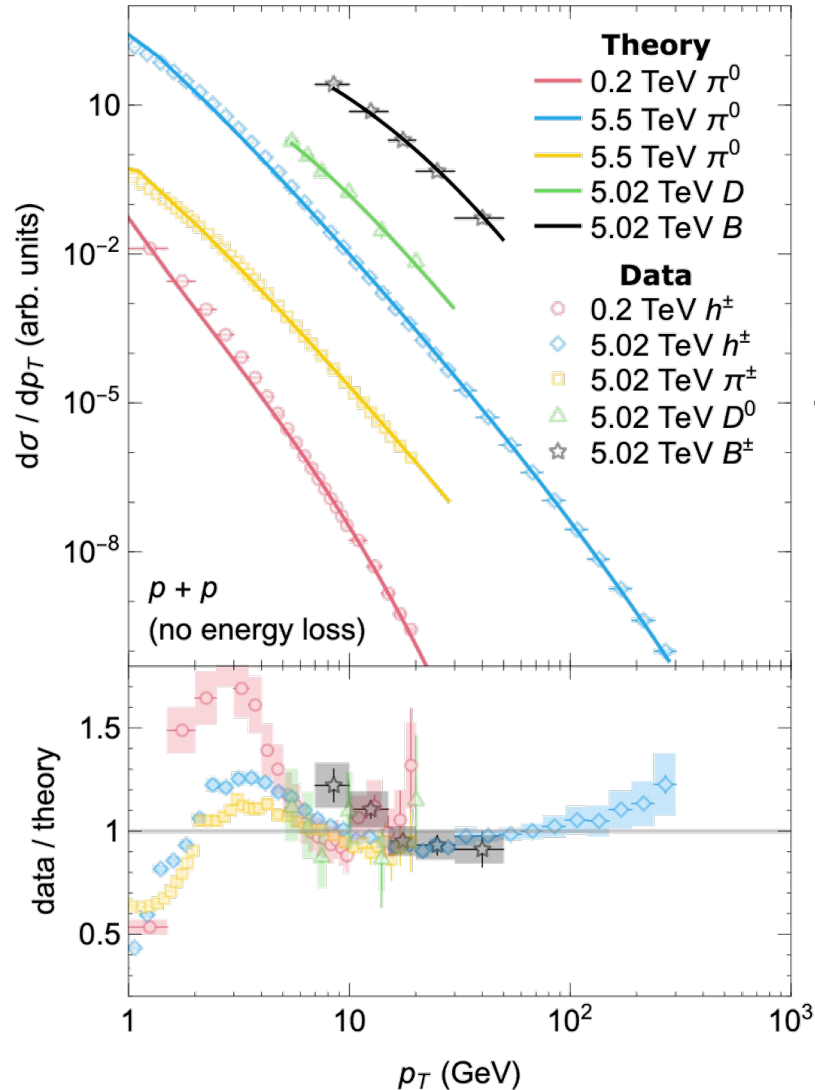
Parton distribution functions (PDFs)

Perturbative QCD cross section for parton interactions

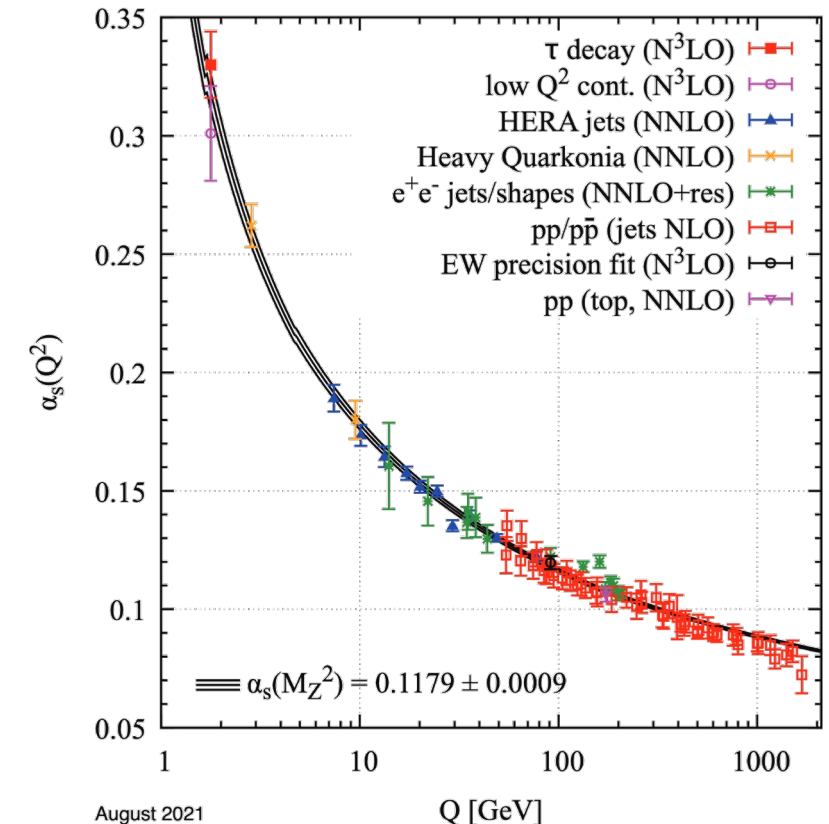
Fragmentation functions

Baseline validation of QCD

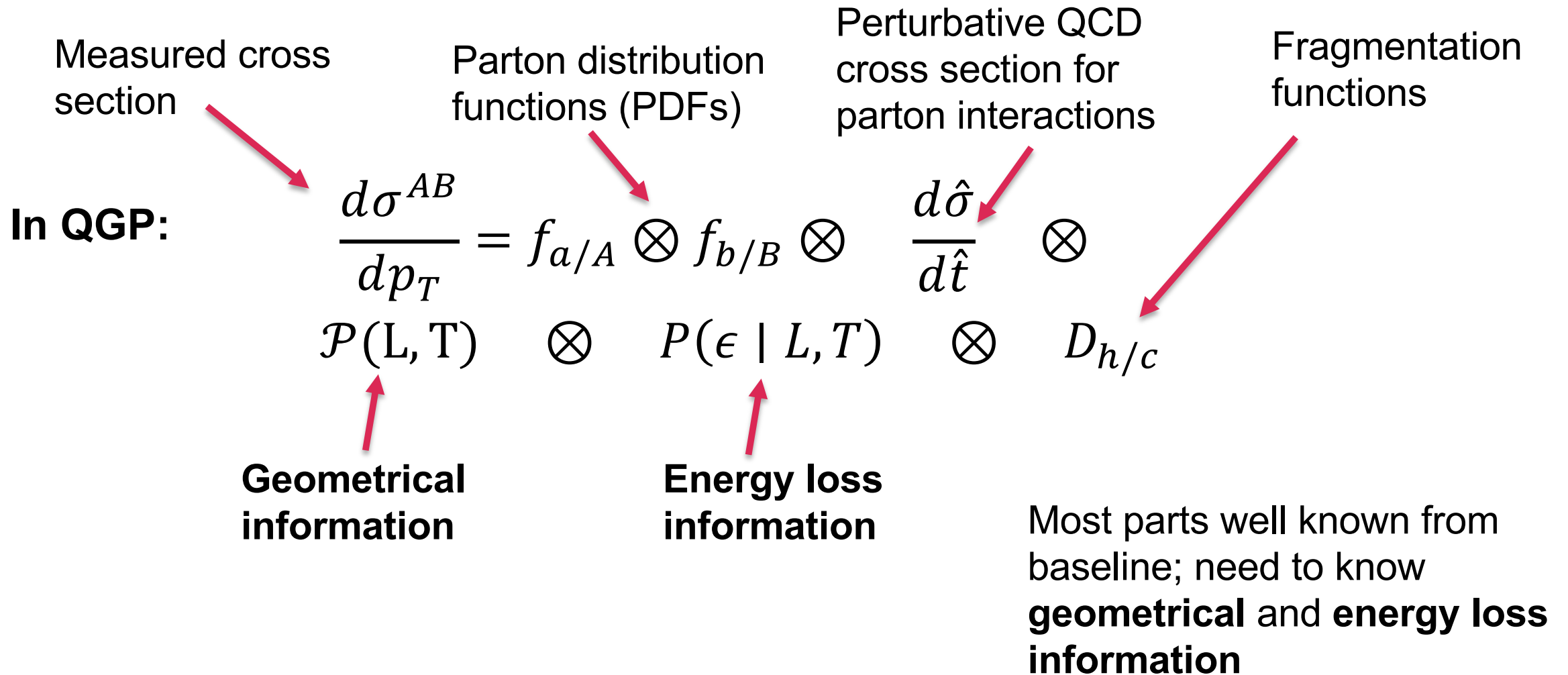
- The coupling gets weaker with energy, as predicted by pQCD



- Even LO pQCD can predict a range of particle spectra very well

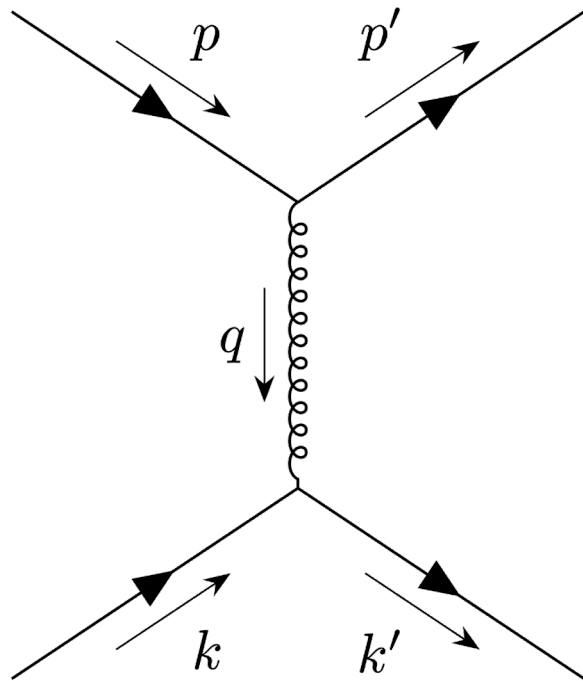


Computing energy loss



> Elastic energy loss computation

Vacuum QCD at asymptotic energies
("Eikonal")



$$\frac{d\sigma}{dt} = \frac{\pi\alpha_s^2}{2} \frac{C_1 C_2}{t^2} \quad t = q^2$$

- α_s is the strong coupling constant
- C_1 and C_2 are the “color casimirs” of the two incoming particles

Based on the first, simple calculation of jet quenching by Bjorken (FERMILAB-PUB-82-059-T)

Estimating the energy loss

- We can build up a simple formula for the energy loss where we already know most of the pieces

$$\frac{dE}{dz} = \sum_a \int d^3k \rho_a(k) \int dt \frac{d\sigma}{dt} v(t)$$

density

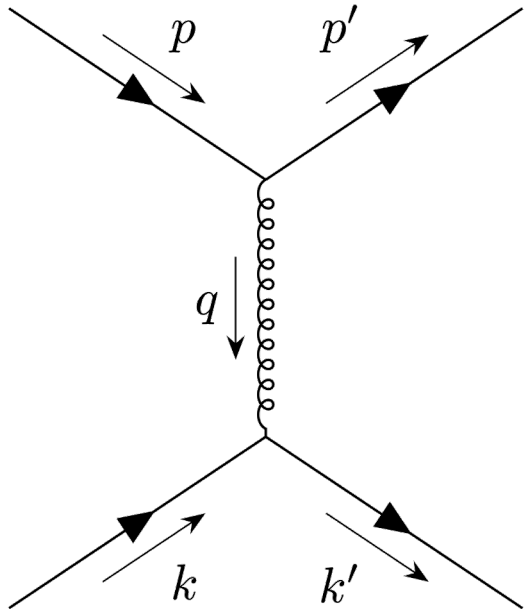
Energy lost

“Probability”
(cross section)
for collision

Energy loss differential
in distance z

Estimating the energy loss

$$\frac{dE}{dz} = \sum_a \int d^3k \rho_a(k) \int dt \frac{d\sigma}{dt} v(t)$$



$$v \equiv E - E'$$

One can show through simple kinematics that

$$v = E \frac{t}{s}$$

For $\rho_a(k)$ we use Bose-Einstein / Fermi Dirac statistics for gluons / quarks

$$\rho_g(k) = \frac{g_g}{(2\pi)^3} \frac{1}{e^{k/T} - 1}, \quad \rho_q(k) = \frac{g_q}{(2\pi)^3} \frac{1}{e^{k/T} + 1}$$

Finally,

$$\int_{|t|_{\min}}^{|t|_{\max}} d|t| \frac{\pi \alpha_s^2 C_R C_a}{2|t|^2} \cdot \frac{|t| E_1}{s}$$

$$= \frac{\pi \alpha_s^2 C_R C_a E_1}{2s} \ln \frac{|t|_{\max}}{|t|_{\min}}$$

$$\frac{dE}{dz} = \sum_a \int d^3k \rho_a(k) \int dt \frac{d\sigma}{dt} v(t)$$

$$= \dots$$

$$= \frac{2}{9} C_R \alpha_s \mu_D^2 \ln \frac{2\langle k \rangle E}{\mu_D^2}, \quad \langle k \rangle \sim 3T$$

- We need estimates for the minimum and maximum momentum transfers.
- **Kinematics** sets the max: $|t|_{\max} = s$
- In principle, $|t|_{\min} = 0$ leads to a divergence, but the QGP gives us a natural $|t|_{\min} = \mu_D^2$, the “**Debye mass**”, the effective mass of the gluon due to screening effects

Radiative energy loss

- Because of interference known as the Landau–Pomeranchuk–Migdal (LPM) effect, radiative does not scale linearly with path length

Elastic

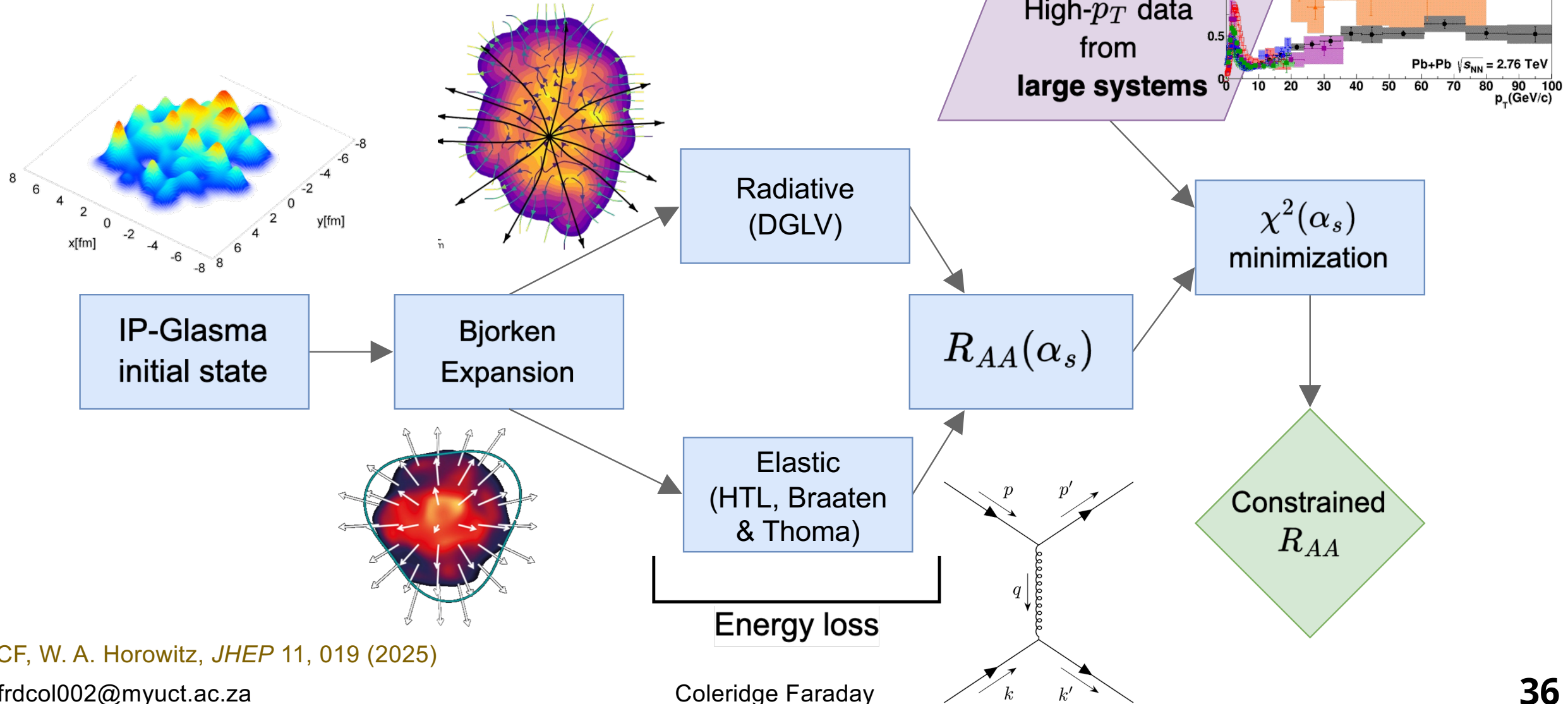
$$\Delta E \sim LT^2 \log \frac{E}{\mu_D}$$

Radiative

$$\Delta E \sim L^2 T^3 \log \frac{E}{\mu_D}$$

In the real model, use a much more complicated and realistic form for both

Putting it all together



CF, W. A. Horowitz, *JHEP* 11, 019 (2025)

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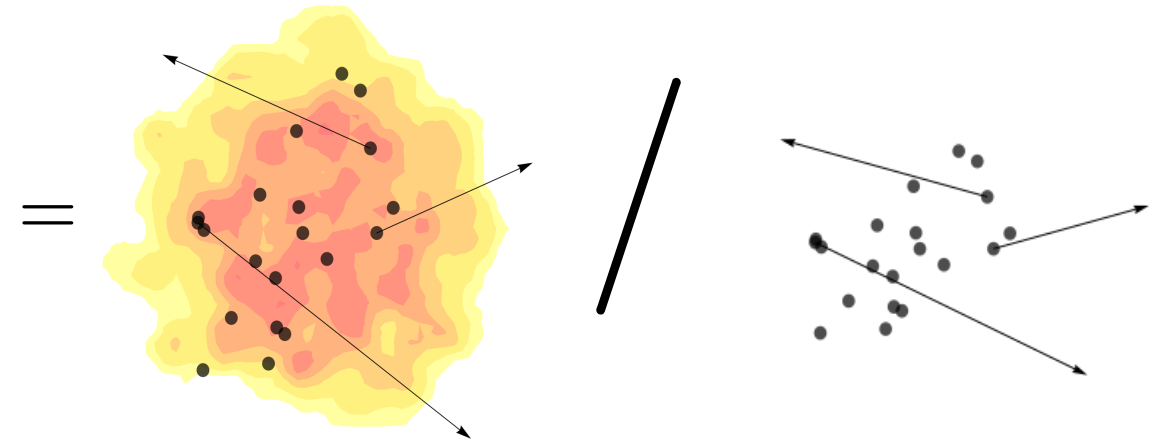
Coleridge Faraday

> Nuclear modification factor

$$R_{AA}(p_T) = \frac{1}{\langle N_{\text{coll}} \rangle} \frac{dN_{AA}/dp_T}{dN_{pp}/dp_T}$$



Glauber model is back!

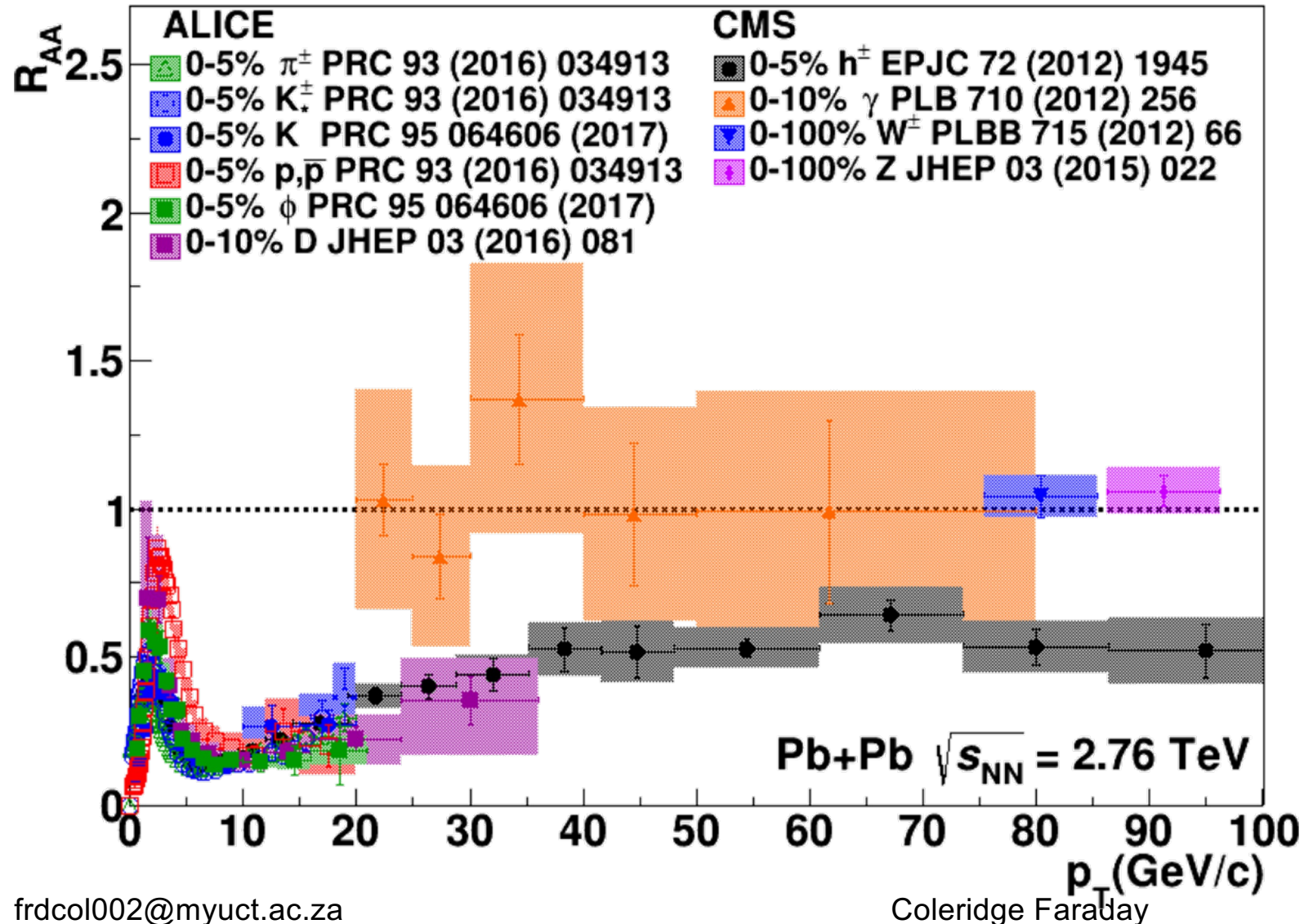


Quantifies how much **energy is lost** in the medium

Initial state, or final state?

- The nuclear modification factor assumes hard yields scale with the number of binary collisions. However,
 - **Nuclear modification of the PDFs**
 - Saturation and shadowing change the hard cross section before anything thermalises
 - **Centrality bias**
 - Centrality is defined by soft multiplicity. If soft and hard production are correlated, selecting on multiplicity biases the hard yield
- Peripheral Pb+Pb shows anomalous suppression that nobody attributes to a medium
- Settling this needs a framework that produces soft and hard particles together — **next week**

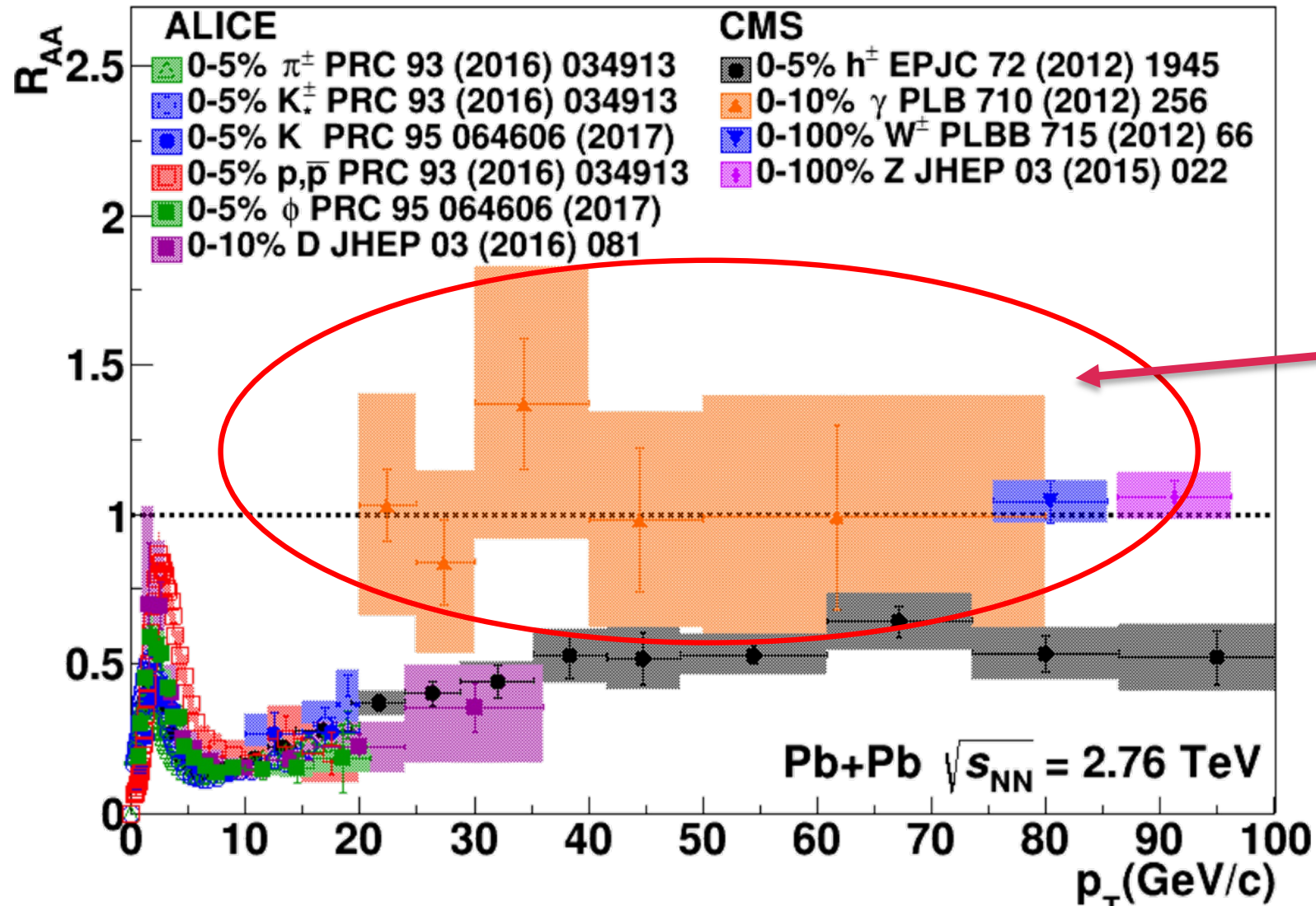
Experimental confirmation



- $R_{AA} \ll 1$ indicates energy loss

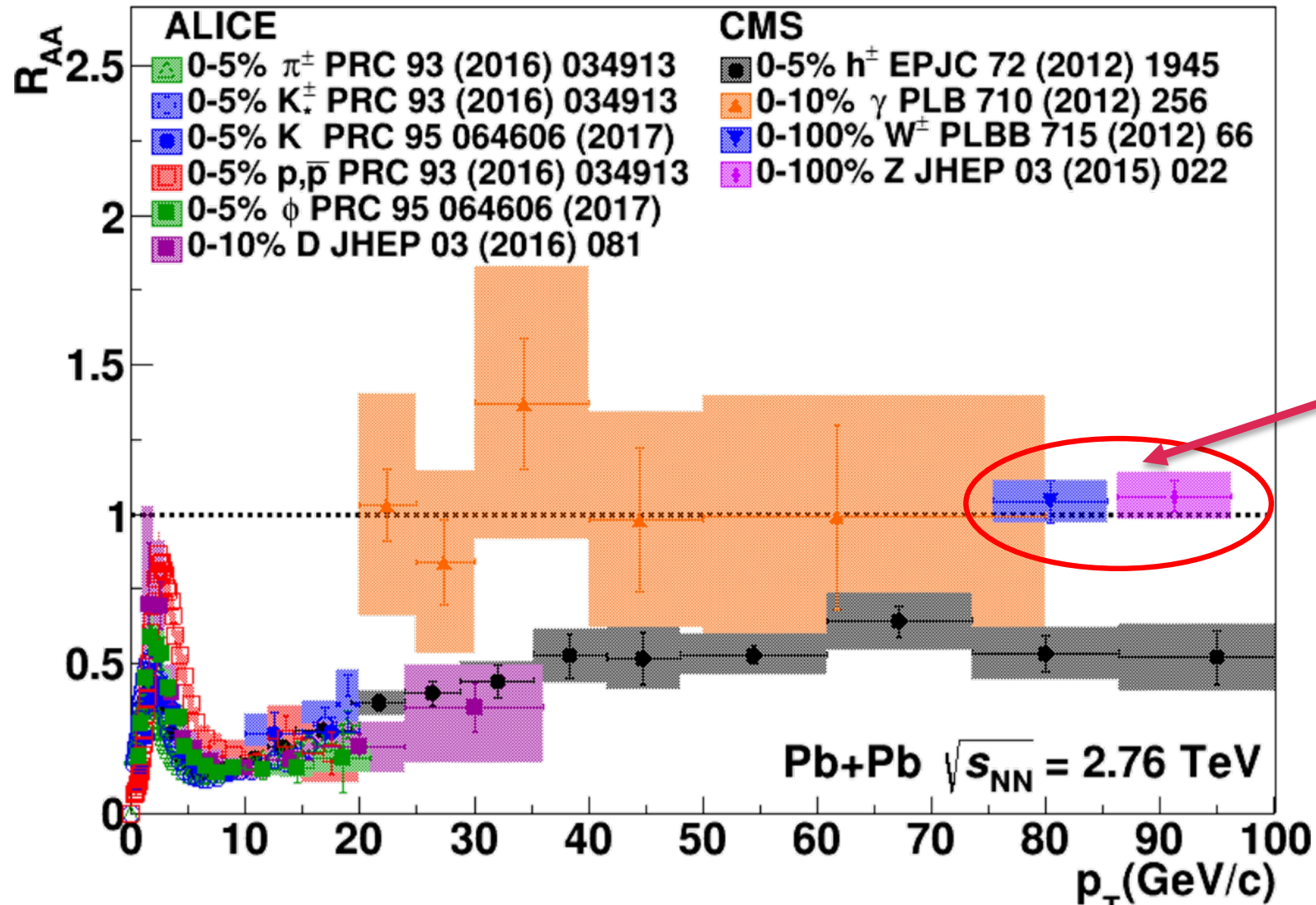
S. Cao and X.-N. Wang, Rept. Prog. Phys. 84, 024301 (2021), arXiv:2002.04028 [hep-ph].

How do we know this is energy loss?



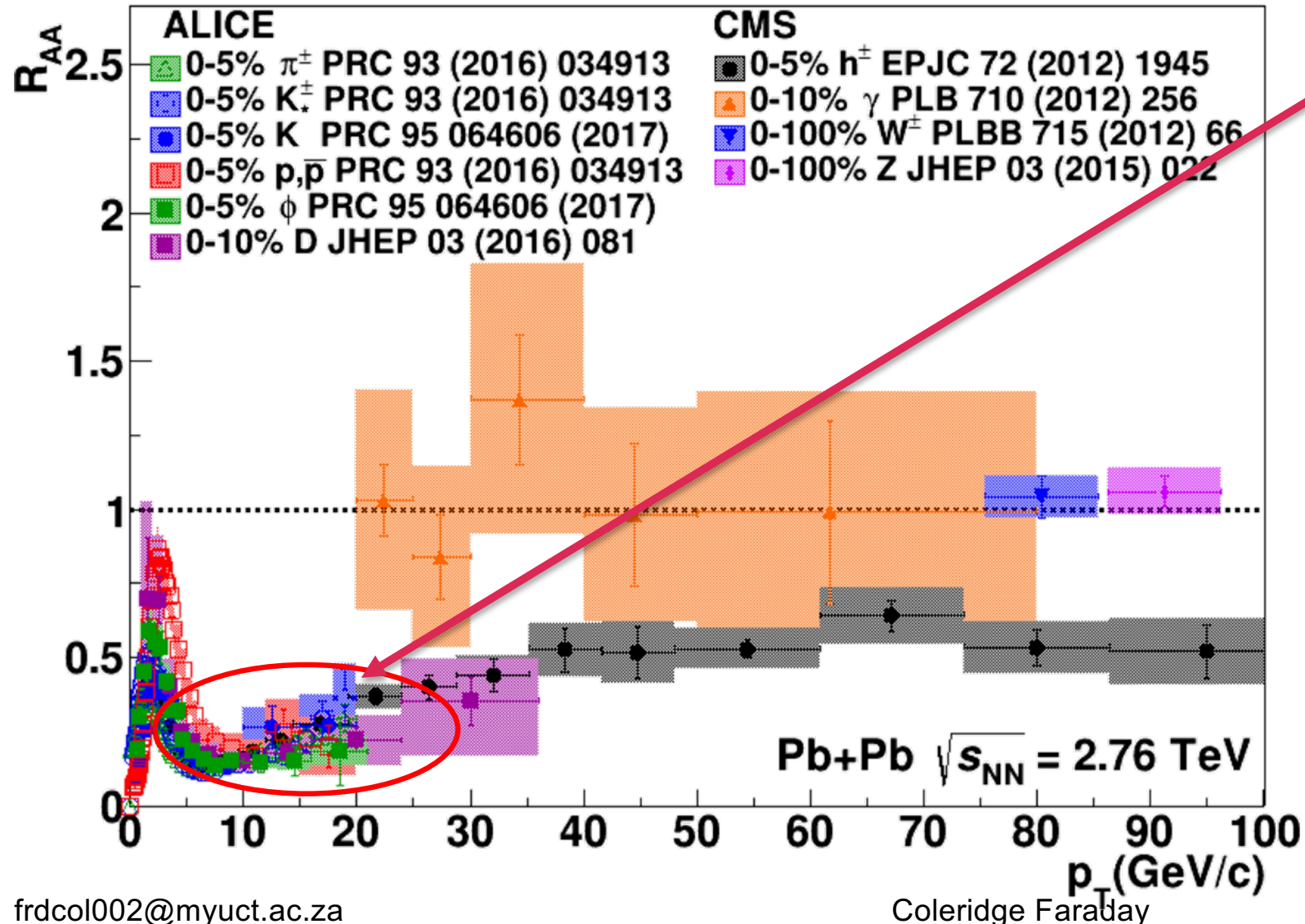
No suppression of photons which do not interact with the QGP

How do we know this is energy loss?



No suppression of W's and Z's which do not interact with the QGP

How do we know this is energy loss?

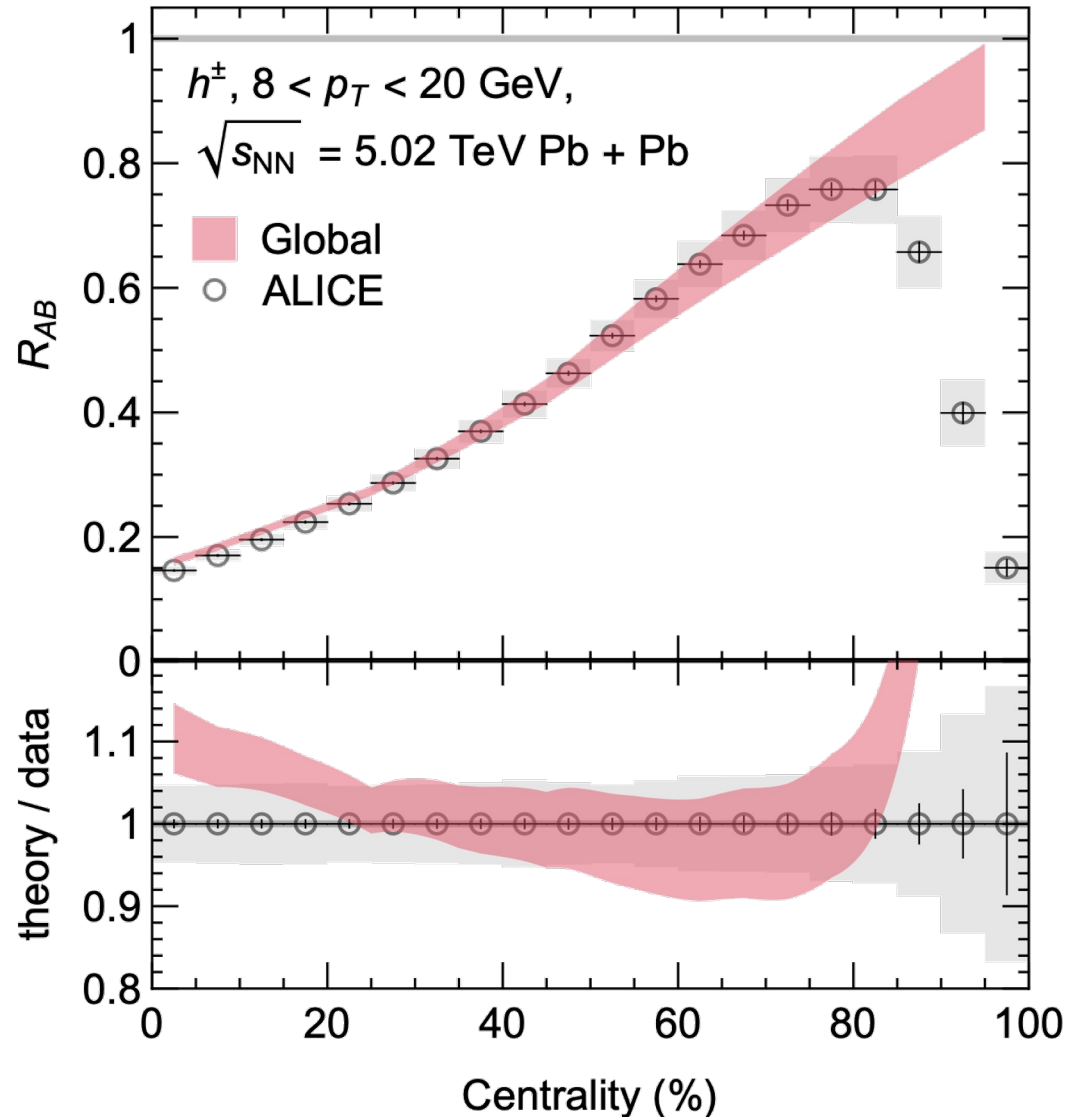


Equal suppression of π , K , and ϕ mesons

- u, d, s are expected to have similar energy loss
 - $\pi = u \bar{u}$, $K = u \bar{s}$,
 $\phi = s \bar{s}$
 - Mass of ϕ is 7 times that of π , so if hadrons lost energy would expect significantly different energy loss
- => Partons lose energy!

S. Cao and X.-N. Wang, Rept. Prog. Phys. 84, 024301 (2021), arXiv:2002.04028 [hep-ph].

How do we know this is energy loss?



- Centrality dependence follows the expectation from energy loss quantitatively and qualitatively
- More central collisions, larger plasmas, more energy loss

CF and W. A. Horowitz, JHEP **11**, 019 (2025).

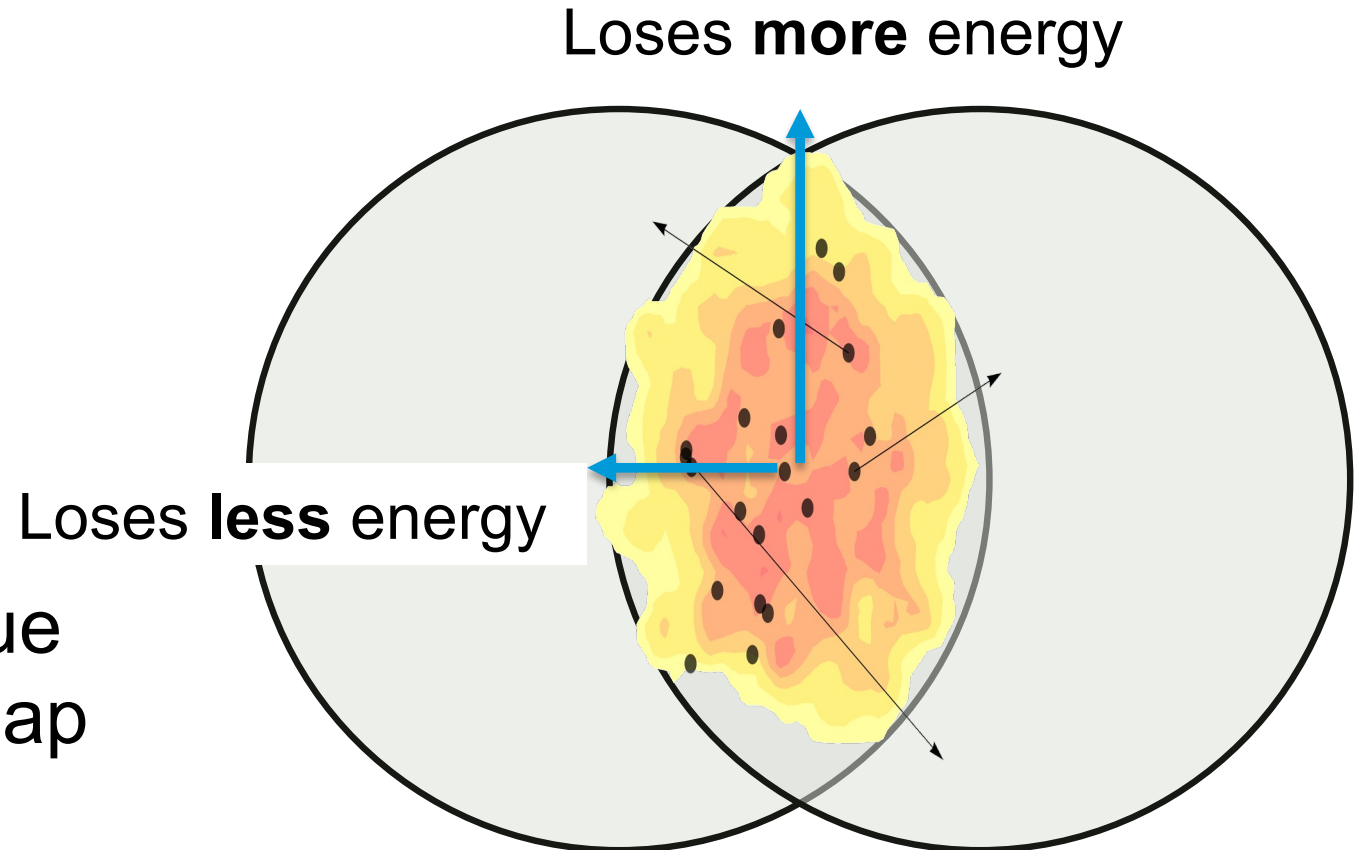
Hard probes of the QGP: high- p_T v_2

Fourier decompose:

$$\frac{R_{AB}(p_T, \phi)}{R_{AB}(p_T)} = 1 + 2 \sum_{n=1}^{\infty} v_n^{\text{hard}}(p_T) \cos \left[n \left(\phi - \psi_n^{\text{hard}}(p_T) \right) \right]$$

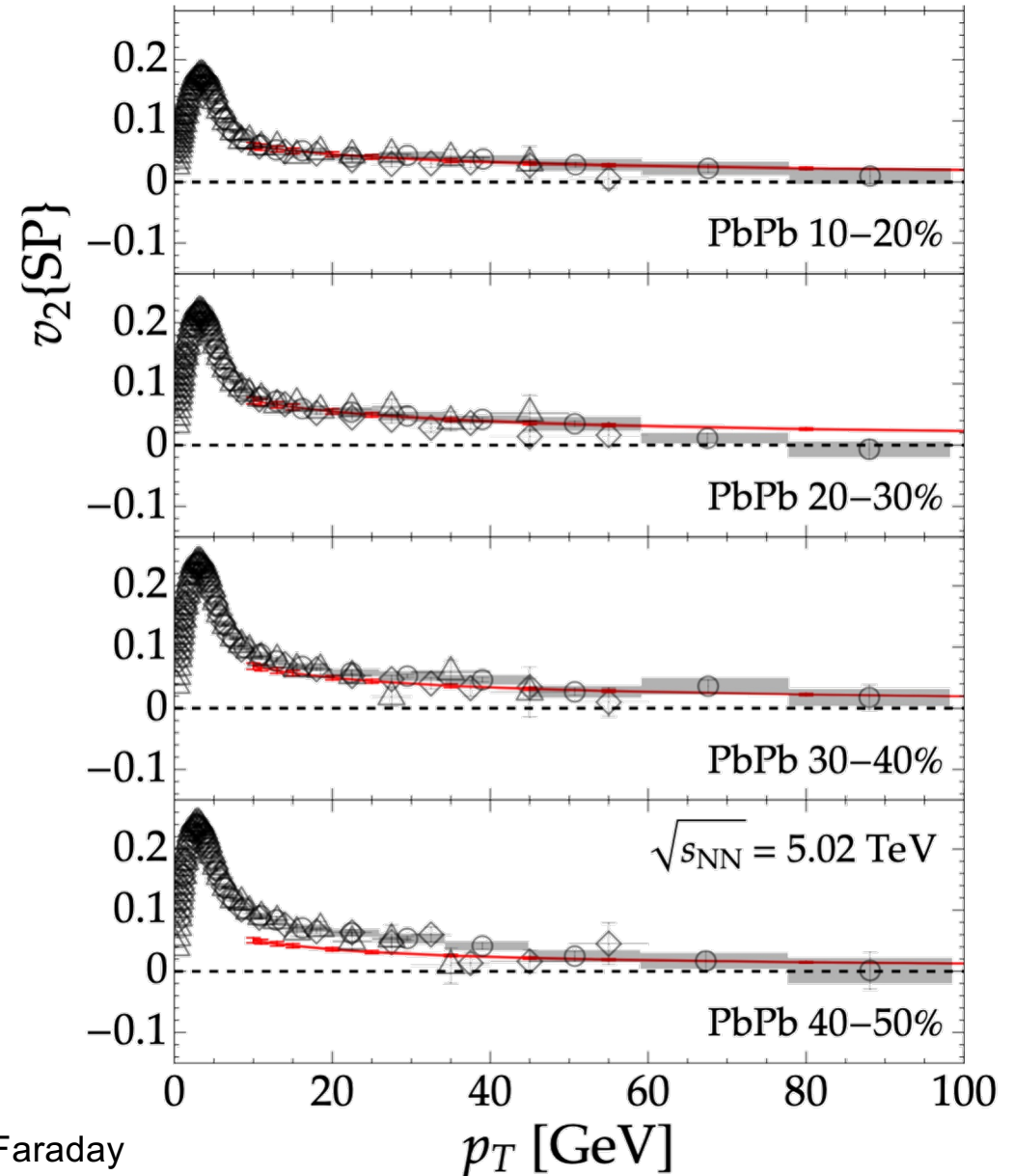
Captures path-length dependent energy loss

v_2 is dominant in PbPb due to elliptical nature of overlap



Experimental confirmation

- Observe large $\sim 0.05 - 0.1$ v_2 at high p_T in data
- Good agreement with theoretical model predictions

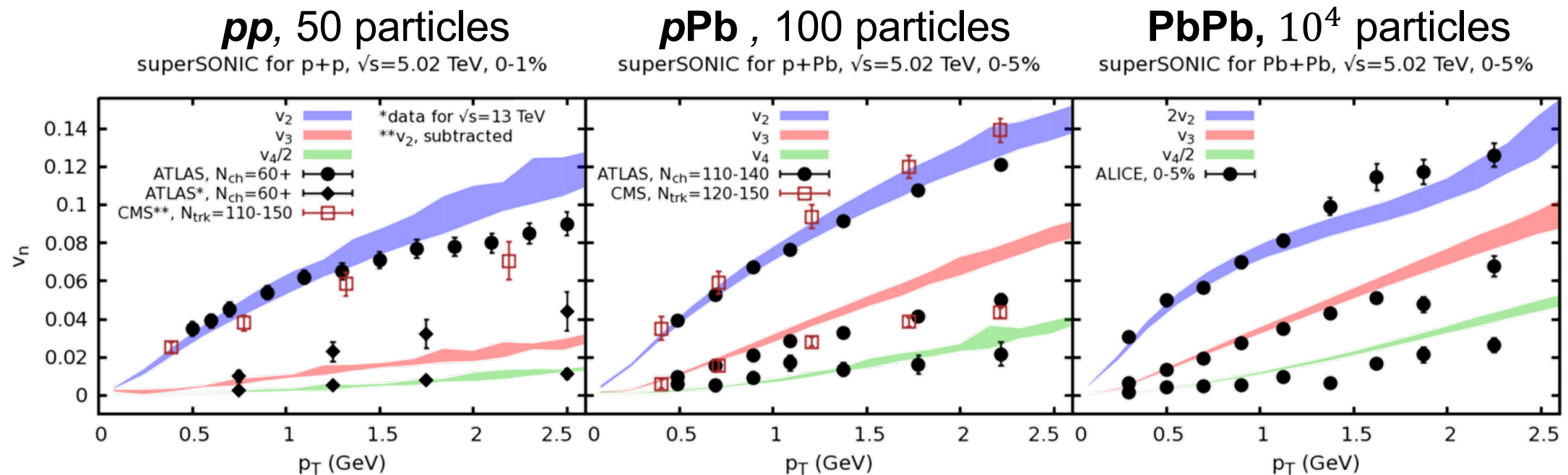




How small can we go?

Collective phenomena in small systems

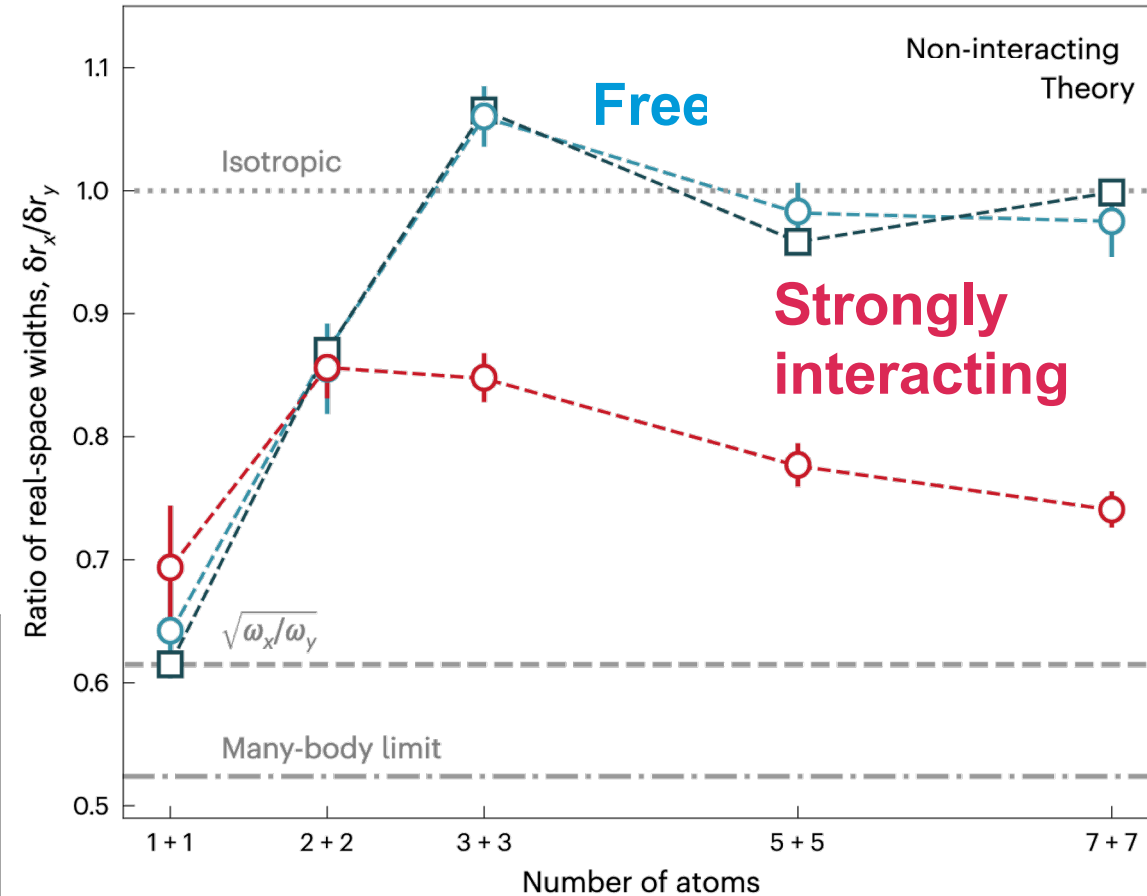
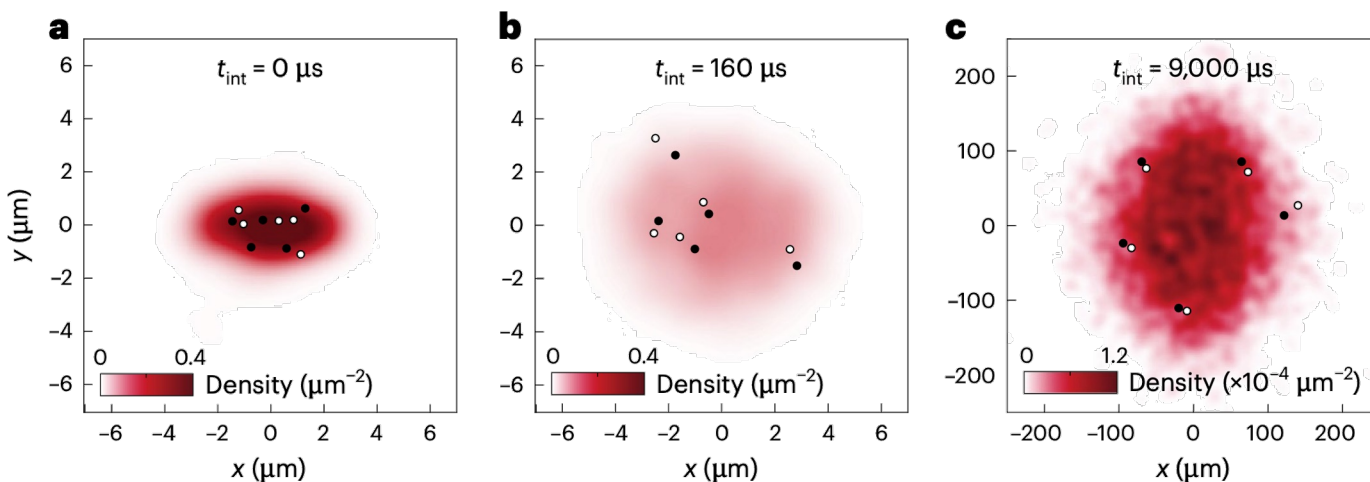
High-multiplicity events in **all collision systems** show **collective phenomena** that hydrodynamics can describe qualitatively.



Weller et al., PLB 774 (2017) 351-356

Surprising lessons from cold atoms

Cold atom experiments say only **six** particles are needed for collectivity!

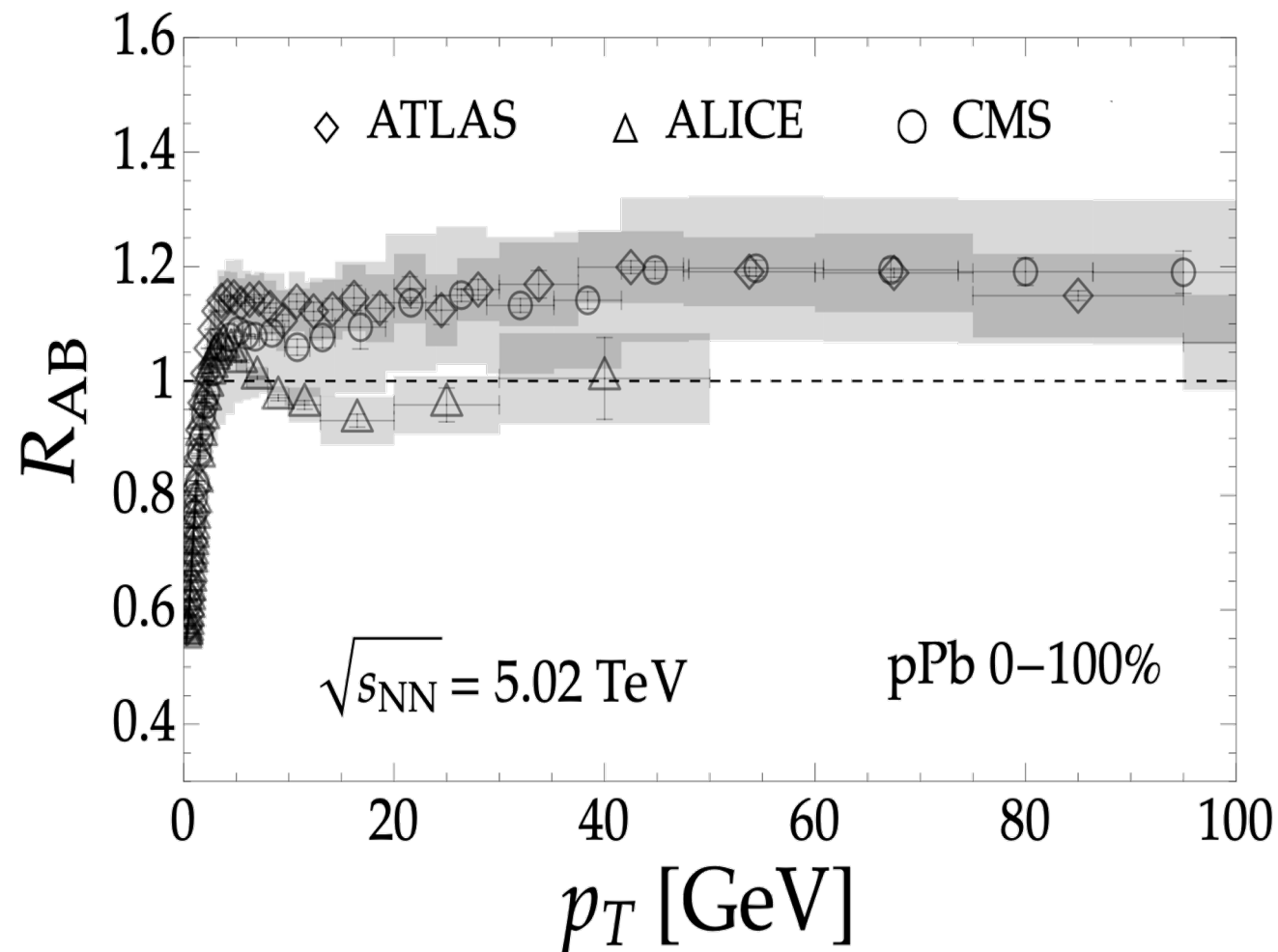


S. Brandstetter, et al., Nat. Phys. **21**, 1 (2025).

Emerging questions

- How many particles are needed before we see collectivity?
- Or more specifically: what size systems can we extrapolate our “standard model of heavy-ion physics” to
- What about energy loss?

Hard probes in small systems

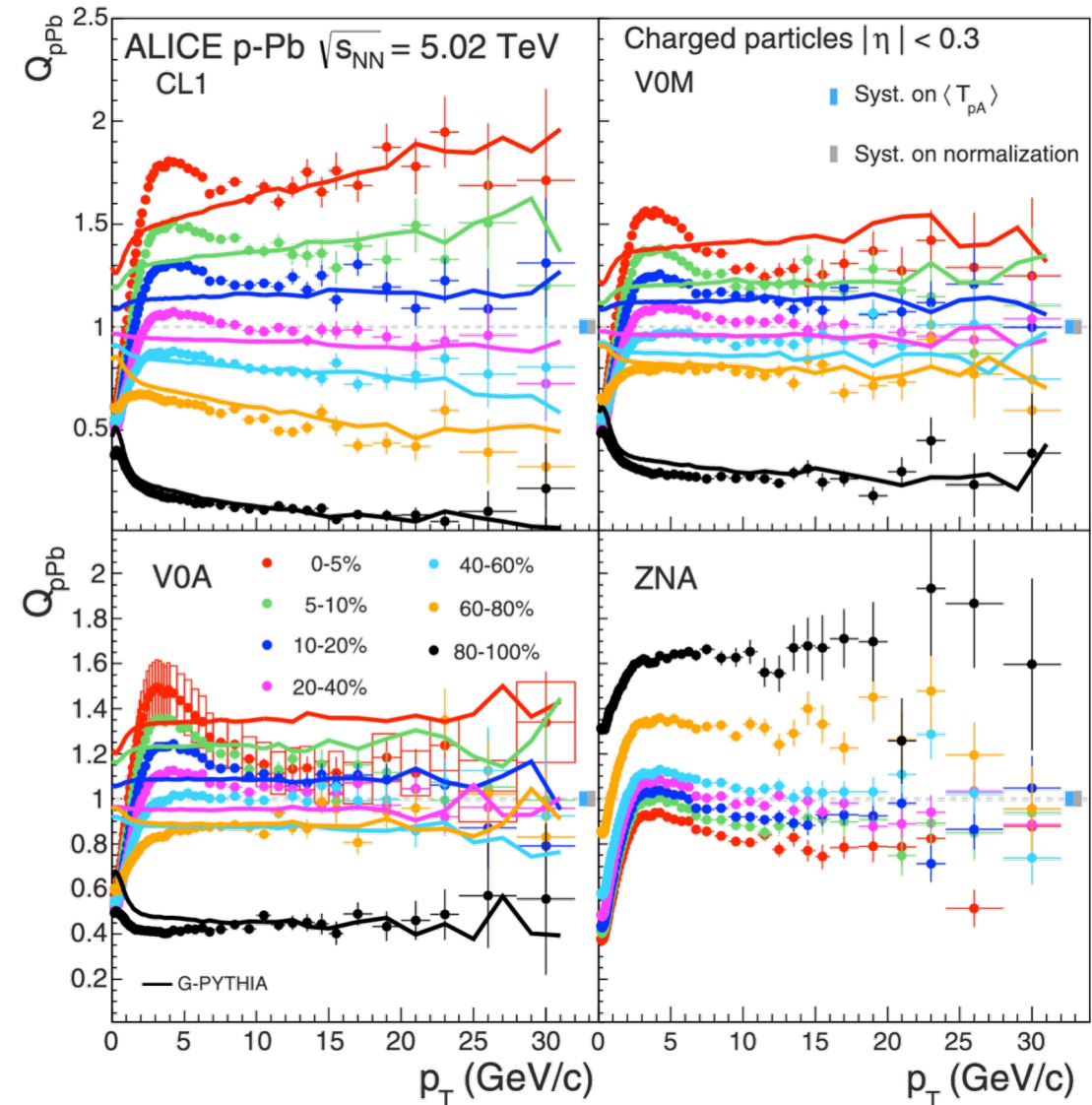


- Nuclear modification factor in minim bias collisions shows no suppression
- Perhaps not enough activity in min. bias, do centrality selection?

Bert, CF, and Horowitz, arXiv:2603.21861

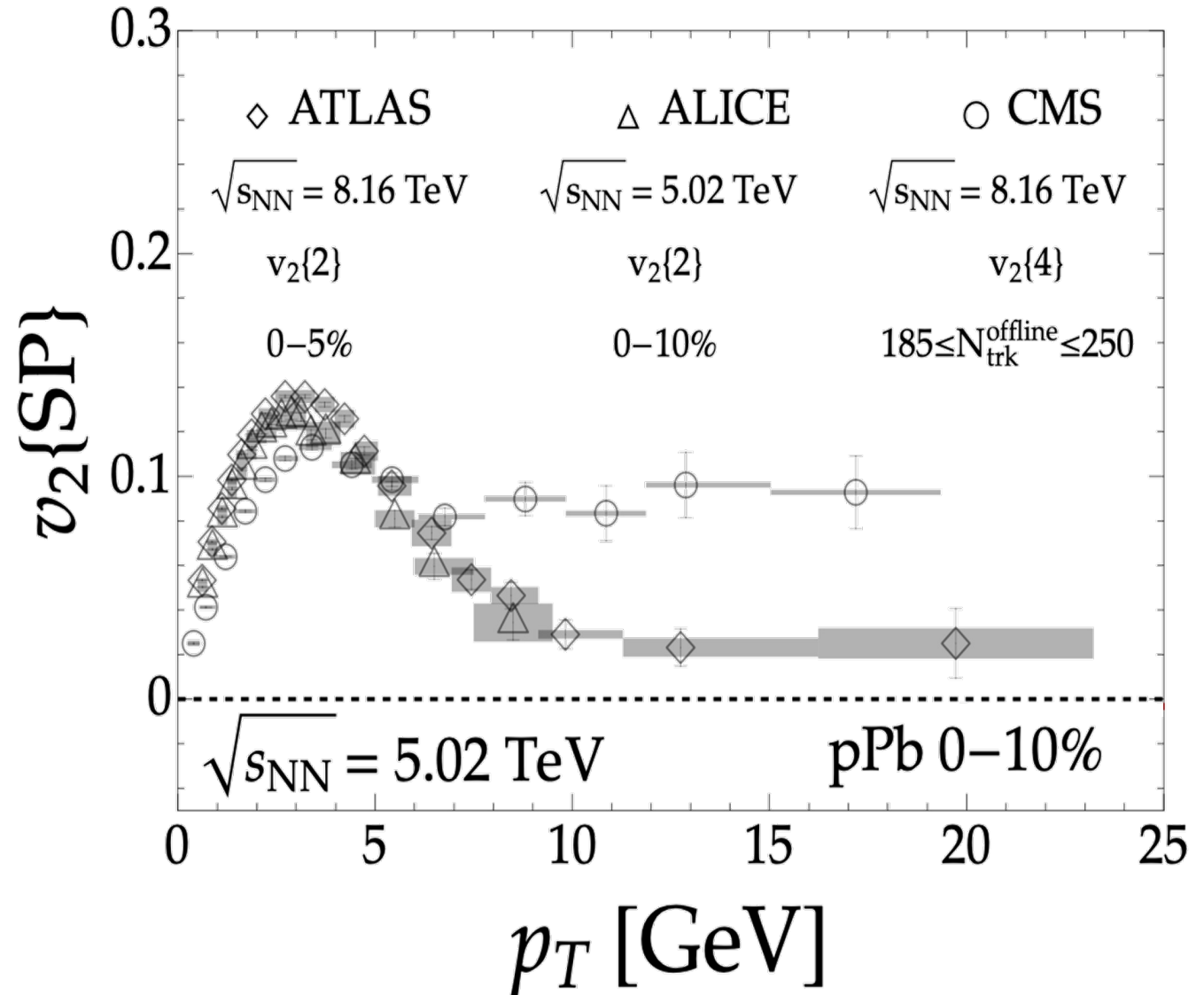
Centrality dependence

- Centrality dependence is even more difficult to understand
 - Strongly depends on centrality definition
 - Enhanced* in central collisions
 - Suppressed* in peripheral collisions
- Can be qualitatively understood as due to hard-soft correlations (more on this next week)



High- p_T v_2 in small systems

- Very large v_2 , similar to Pb + Pb $\Rightarrow$ energy loss?



Hard probes in small systems

- Broadly, mixed messages from hard probes in small systems
- No clear picture of energy loss or no energy loss
- **Is a plasma forming in small systems?**

Observable or effect	Pb- $A + A$	p-I $p + A$	pp $p + p$	Refs.
Near-side ridge yields	✓	✓	✓	1-8,8,9
Angular anisotropy	✓ v_1-v_9	✓ v_1-v_5	✓ v_2-v_4	2-8,10-25
Flow	✓	✓	✓	16,26-34
ss dependence	✓ v_2-v_5	✓ v_2, v_3	✓ v_2	12,15,17,21,35-42
Higher-order cumulants (mainly $v_2\{n\}, n \geq 4$)	✓ “4 $\approx$ 6 $\approx$ 8 $\approx$ LYZ” +higher harmonics	✓ “4 $\approx$ 6 $\approx$ 8 $\approx$ LYZ” +higher harmonics	✓ “4 $\approx$ 6”	17,18,23,32,41,43-55
Symmetric cumulants (SC)	✓ up to (5, 3)	✓ only (4, 2), (3, 2)	✓ only (4, 2), (3, 2)	20,23,56-62
Non-linear flow modes	✓ up to v_7	?	?	24,63,64
Correlations	✓ $n = 2-4, \{2\}, \{4\}$	✓ $n = 2, 3, \{2\}$	?	7,19,65-69
	✓ v_2-v_4	?	?	70-72
	✓ up to v_4	✓ v_2	?	73,74
Directed flow (from spectators)	✓	?	?	75
Charge-dependent correlations	✓	✓	✓	76-82
Freezeout properties	✓	✓	✓	83-94
	✓	✓	✓	85,88,92,95-99
	✓ GC level $\gamma_s^{\text{GC}} = 1$	✓ GC level $\gamma_s^{\text{GC}} \approx 1$	✓ GC level $\gamma_s^{\text{C}} < 1$	85,86,89,90,100,101 93,94,102-104
	$R_{\text{out}}/R_{\text{side}} \approx 1$	$R_{\text{out}}/R_{\text{side}} \lesssim 1$	$R_{\text{out}}/R_{\text{side}} \lesssim 1$	105-113
Direct photons at low p_T	✓	?	✗	114-116
v_n in events with Z, jets	?	✓ up to v_3	✓ v_2	117-119
Hard probes	✓ v_2	✓ v_2	✓ v_2 in jet frame	120,121
	✓	✗	✗	122-129
	?	✓	?	?
... through dijet asymmetry	✓	✗	✗	140-147
... through correlations	✓ (Z-jet, γ -jet, h-jet)	✗ (h-jet, jet-h)	?	139,148-157
Heavy-flavor	✓	✓	?	117,158-160
	✓ up to v_3 (c), up to v_2 (b)	✓ up to v_2	✓ up to v_2	40,161-183
	✓	✓	?	167,184-219

✓ = observed
✗ = not observed
? = not measured

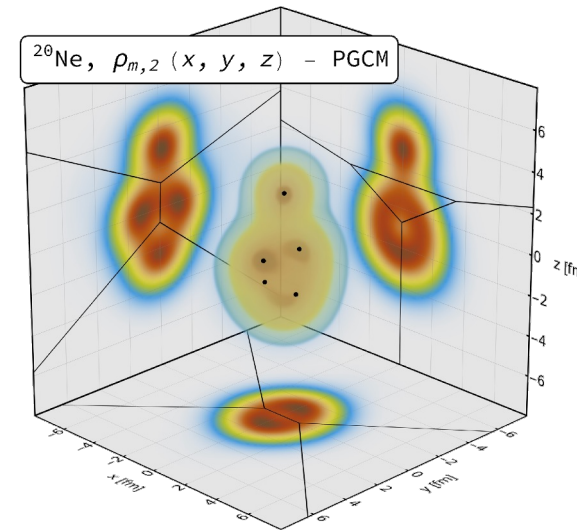
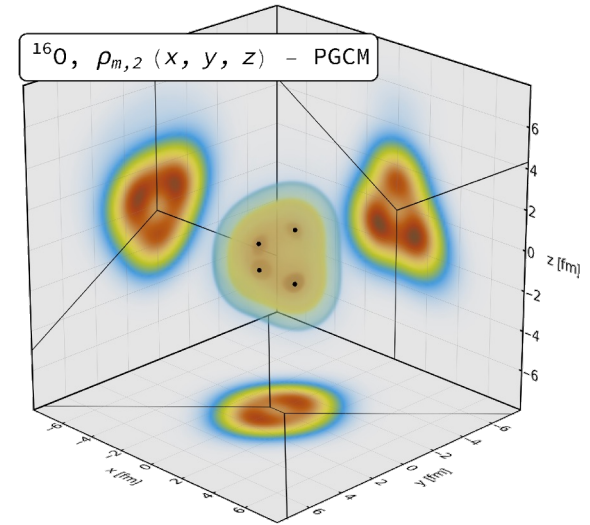
Mixed messages?

Hard probes are *the* missing piece of the puzzle!

Adapted from J. F. Grosse-Oetringhaus and U. A. Wiedemann, arXiv:2407.07484 [hep-ex] (2024). See backup slides for references

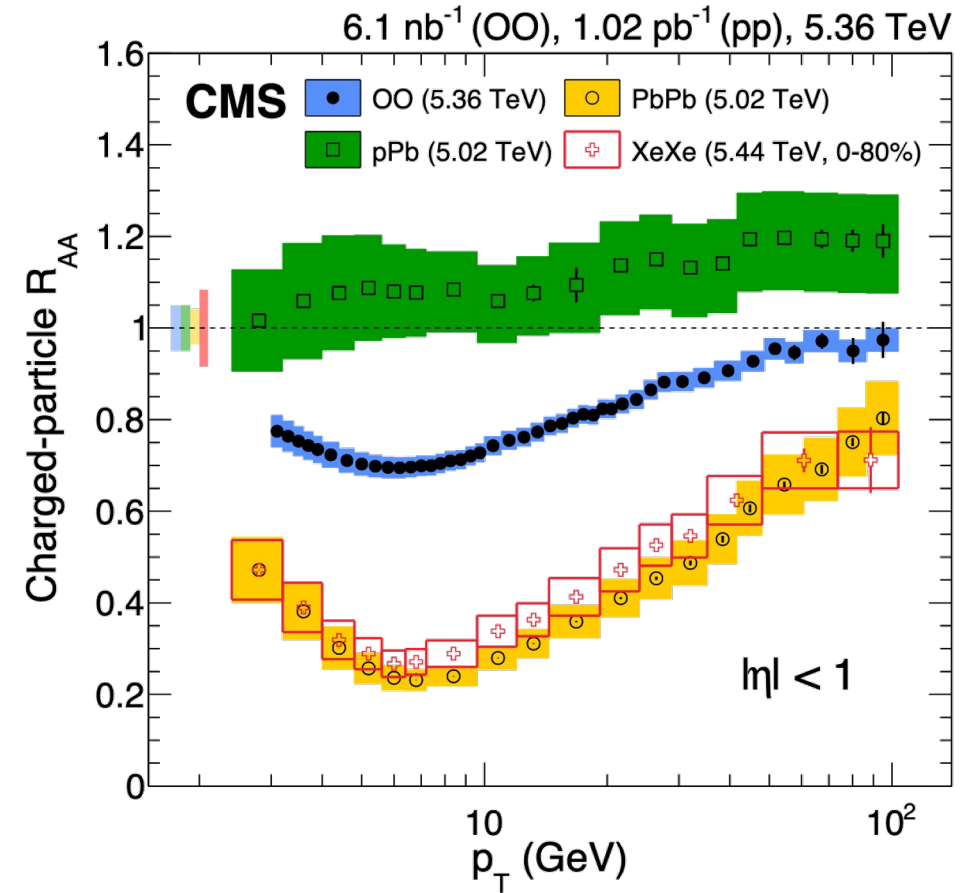
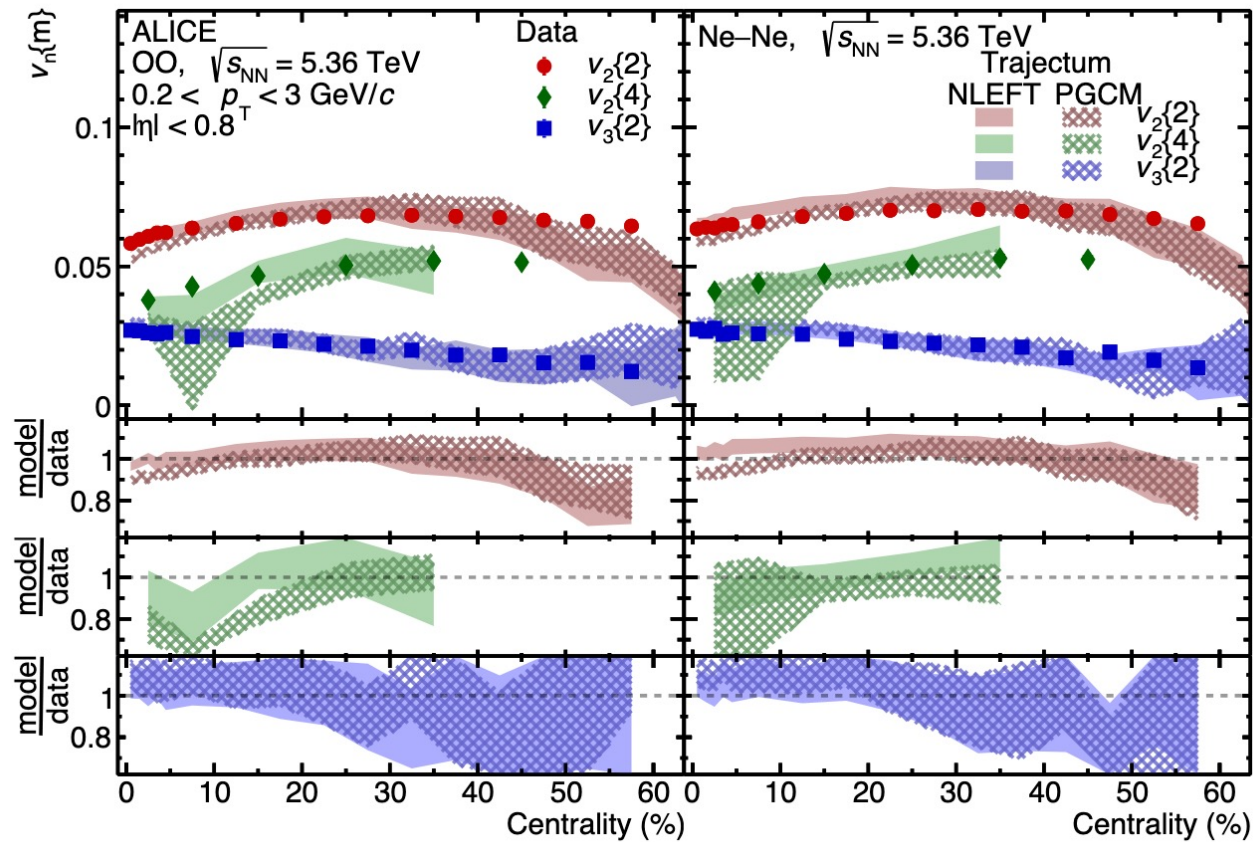
Light-ion collisions

- Collisions of **oxygen** and **neon** are an ideal testing ground for small system energy loss
 - Symmetric, meaning 3D hydro not mandatory
 - No centrality-cuts needed
 - Large enough to expect non-trivial energy loss signal
 - Large enough to expect hydro to apply
 - Similar multiplicity to central p + Pb



Giacalone, G. *et al. Phys. Rev. Lett.* 135, 012302 (2025)

Flow + energy loss in light ions

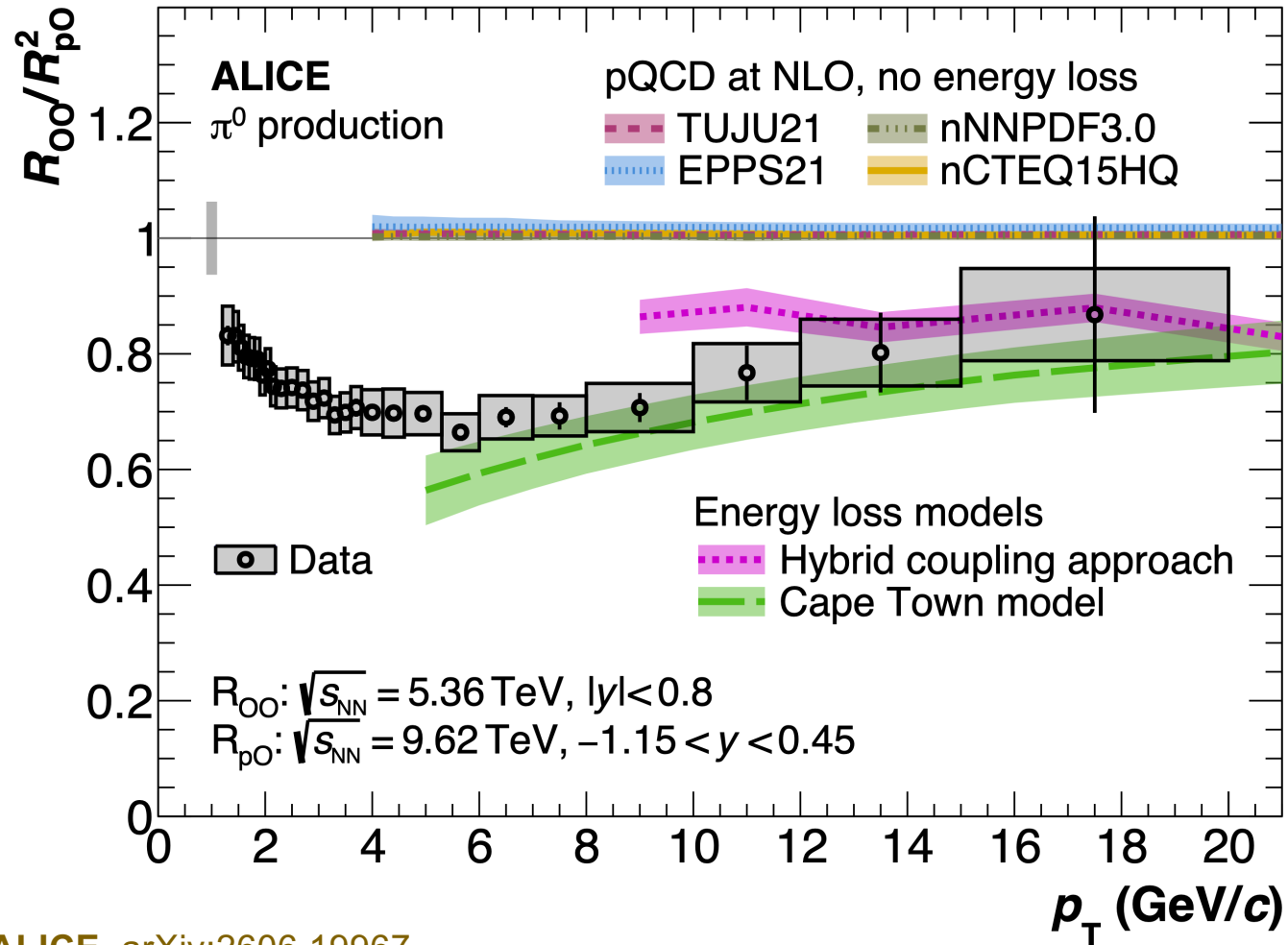


ALICE arXiv:2509.06428 [nucl-ex] (2025)

CMS arXiv:2510.09864 [nucl-ex] (2025)

→ The heavy-ion paradigm extends to (at least qualitatively) OO and NeNe collisions!

Predictions for oxygen



- Good agreement between (blind) predictions and data!
- Next week: more on how we get these results and on extending this program to lighter ions

ALICE, arXiv:2606.19967

CF, W. A. Horowitz, *JHEP* 11, 019 (2025), arXiv: 2505.14568

The current state

- The **small system energy loss puzzle** has been partially resolved by light ions. We can
 - Understand both collective flow etc. and jet quenching in light-ion collisions, which are significantly smaller than Pb+Pb collisions
 - Blind model predictions in quantitative agreement with data
- However, **several open questions remain**:
 - How can we understand the large collective flow but lack of jet quenching in high multiplicity $p + \text{Pb}$ and $p + p$ collisions?
 - How small can we extend the heavy-ion paradigm?

No conclusions yet...

Next time:

- What one can learn from fits to experimental data
- Theoretical framework that includes correlations between hard and soft particles from saturation physics
 - Potential explanation for the strange p Pb suppression data?
- Quantitative proposal for even-lighter-ion collisions in future LHC runs with ions as small as helium.